CLIPPED GRAM ENERGIES AND ZERO COUNTS OF THE RIEMANN ZETA FUNCTION

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Abstract. We study the interaction energy that remains available when the Gram matrix associated with the zeros of the Riemann zeta function is spectrally clipped. An explicit band-limited majorant and a close-pair estimate transfer an existing eight-point inequality to the full clipped energy, uniformly over configurations of arbitrary size. Together with the unconditional pair-correlation formula this gives

$$\liminf_{T\to\infty} \frac{N_{\mathrm{s}0}(T)}{N(T)} \geq \frac{66812491}{99194876} = 0.673547805029767868\ldots,$$

where $N_{\mathrm{s}0}$ counts simple critical-line zeros and N counts all nontrivial zeros with multiplicity, giving 67.35478050%. The corresponding average of the simple-zero and critical-line proportions is at least 83.67739025%. Evaluating separate labelled and multiplicity certificates gives 83.90046180% for distinct locations, 67.80092360% for simple zeros anywhere, 88.93165002% for simple-or-critical-line mass, and 83.79775462% for mass at locations of multiplicity at most two; such locations constitute at least 93.94524765% of the distinct locations. All terminating lower percentages are rounded down. Beyond these numerical deductions, we prove a phase comparison in which local slack pays continuously for long physical spans, derive a common residual for joint profile-and-threshold deductions, and solve a two-lattice variational problem explicitly. A twelve-fibre construction extends a periodic-mixture obstruction to the full affine L^2 space and gives a 67.404% ceiling for the specified joint-affine simple-line relaxation. A simultaneous union ceiling is 89.497833220444%. These ceilings describe limitations of the relaxation, not upper bounds on the actual zero proportions.

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1. Introduction

A multiple zero is expensive in a quadratic energy: multiplicity j contributes j to the zero count but j^2 to the diagonal energy. Nearby distinct zeros also interact. A lower bound for this interaction leaves less of the available energy for multiplicities and for zeros away from the critical line. The central issue is to retain that interaction through the spectral inequality which makes the argument unconditional.

For a nonnegative even density f of integral one on $I = [-1/2, 1/2]$, let K_f denote its Fourier transform. On the critical line, normalized zero locations y_i determine unit vectors whose Gram matrix has entries $K_f(y_i - y_j)$. Its off-diagonal square energy measures the interaction. Off the critical line, reflection symmetry gives Hermitian terms with at most one positive direction per reflected pair. It does not give positive rank-one terms. The appropriate comparison therefore uses a clipped square

$$\phi_c(x) = x^2 - (x - c)_+^2$$

rather than the full square on an arbitrary retained Gram matrix. Recovering full energy where it is legitimate, and keeping track of what is lost elsewhere, will be the organizing principle of the paper.

The analytic foundation is the unconditional Montgomery theorem of Baluyot, Goldston, Suriajaya and Turnage-Butterbaugh [1]. It evaluates a pair sum over the complex zeros without first placing them on the critical line. Their subsequent work [20] studies the horizontal information available

from pair correlation under a narrow-strip hypothesis. The unconditional finite-rank and Hilbert-space arguments of Alpöge–Furman and Lamzouri [2, 3] handle the off-line pairs by their inertia instead. With the Montgomery–Taylor profile they yield more than 67.25% simple critical-line zeros. These arguments descend from Montgomery's pair-correlation method and its test-function optimization [21, 22]; conditional semidefinite refinements appear in [23]. The present argument uses the unconditional complex-zero formula throughout.

Local Gram inequalities have sharpened the basic diagonal estimate. The relevant developments include overlapping finite windows, pressure optimization, bounded correction potentials and compatible window systems [5, 6, 9, 4, 7, 8]. Close-pair separation allows the full energy of a separated remainder to survive clipping [11, 12, 10]. Labelled inequalities distinguish simple and double locations [19, 24, 10], while residual and multiplicity accounting support further consequences [3, 4, 16, 15, 26]. Related numerical refinements and their public certificate data are reported in [14, 17, 18]. These ingredients have different roles. A stronger finite inequality is useful only when its Fourier profile, its trace energy and the clipped operator to which it is applied remain the same.

Our principal numerical result is

$$\liminf_{T\to\infty} \frac{N_{\mathrm{s}0}(T)}{N(T)} \geq q_* := \frac{66812491}{99194876} = 0.673547805029767868\ldots. \tag{1.1}$$

Here $N(T)$ counts nontrivial zeros with $0 < \Im\rho \leq T$, with analytic multiplicity, and $N_{\mathrm{s}0}(T)$ counts those that are simple and lie on $\Re\rho = 1/2$. The numerical profile and its eight-point floor are specified public inputs [6, 25]. For that profile we give an elementary explicit majorant with Fourier support $[-17/20, 17/20]$ and integral strictly below two, together with a close-pair kernel lower bound. Every separated Gram matrix is then below the clipping level, regardless of its size. A maximal matching of close pairs handles the complement. Summing the local inequality on the separated remainder uses the full lag capacities and incurs only a fixed endpoint loss. The smoothing needed by the analytic formula costs $O(\delta n)$, not $O(\delta n^2)$. These facts turn the local floor into the global estimate

$$\operatorname{tr}\phi_2(U) - n \geq \frac{805124}{10^8} n - \frac{404350}{10^8} S - O(1),$$

where S is the span and U is the normalized Gram matrix. This realization of an established separation mechanism, for the specified profile with explicit uniform constants, yields (1.1).

Keeping the positive-index information also gives

$$\liminf_{T\to\infty} \frac{N_{\mathrm{s}}(T) + N_0(T)}{2N(T)} \geq \frac{1+q_*}{2} = 0.836773902514883934\ldots.$$

The two counts in this average have different meanings: N_{s} counts simple zeros anywhere, whereas N_0 counts all critical-line multiplicity. They are not each bounded below by the displayed average. Table 1 collects seven consequences, with every numerator and denominator identified. The remaining five values come from evaluating other stated certificates at sharper exact energy bounds, not from adding the gains of incompatible profiles. The small numerical refinements are kept distinct from the structural results.

The first structural question concerns information left after a local inequality is summed. Its nonnegative slack H and the energy T at unused index lags are often both discarded. A sinc-type lower bound for the Fourier kernel converts small long-lag energy into phase agreement. The difficulty is that a fixed index lag can have an arbitrarily large physical span. We show that the sum of the local slacks over a frame cover pays for that span continuously. The result is a uniform estimate for the squared distance of adjacent gaps to the integers. For one explicit profile,

$$\sum_i \operatorname{dist}(y_{i+1} - y_i, \mathbb{Z})^2 \leq 350(H+T) + 6. \tag{1.2}$$

More generally, under fixed local slack and endpoint hypotheses, a degree-D cosine profile permits an $O(D)$ energy price. The argument keeps a geometric residual which an affine count alone does not record. Its profile is separate from that of (1.1).

The complementary structural question is how much any deduction from a specified collection of affine floors can establish. Periodic configurations supply tests of every universal floor. We put their profile dependence, the clipping thresholds and the unused test slack into one exact residual. The residual permits arbitrary fixed finite combinations of profiles, with separate thresholds, while testing all of them on common configurations. It leads to two variational conclusions. A mixture of the unit lattice and the spacing-two lattice has an explicit positive minimizer, a coercive error term and an exact two-count frontier within that two-test relaxation. A period-three/period-four mixture, originating in the finite-dimensional obstruction of [13], admits a twelve-fibre formulation on all of $L^2(I)$. A rational approximate minimizer and a rigorous residual bound yield the 67.404% ceiling stated in the abstract. The two-lattice frontier also gives the simultaneous 89.497833220444% union ceiling.

A ceiling here is a feasibility statement about retained inequalities. It says that those inequalities allow a particular abstract count population; it does not construct zeros with those counts. This distinction is productive: it identifies the extra information a stronger deduction must retain. The phase comparison retains such information, as do labelled-block constraints not included in the specified relaxation. In a smaller scalar multiplicity relaxation we can be more explicit: a population of simple and triple real locations satisfies every scalar clipping row and simultaneously attains the two cutoff-two bounds. Changing the clipping level alone cannot improve these bounds.

We first establish the signed-operator interface and prove the majorant and full-energy transfer for the main profile. We then give the count deductions, the phase comparison, and the joint variational analysis. The appendices specify the finite rational calculations and the data used in the twelve-fibre estimate. Computer-assisted local inequalities are cited with their precise profiles and domains; all subsequent operator, counting and variational arguments are given here. The accompanying formal source covers the main numerical simple-line chain, while the additional count and variational results have the proofs and arithmetic certificates specified in the paper.

2. One operator and the counting interface

2.1. **Profiles, complex zeros, and the trace energy.** Put $I = [-1/2, 1/2]$. For an even probability density $f \in L^2(I)$, extended by zero, define

$$K_f(t) = \int_I f(u) e^{2\pi i t u}\,\mathrm{d}u, \qquad R(f) = \int_I f^2 + \iint_{I^2} |s-t| f(s) f(t)\,\mathrm{d}s\,\mathrm{d}t. \tag{2.1}$$

The same integral defines $K_f(t)$ for complex t and is entire, because f is integrable with compact support. We use $\operatorname{sinc}(t) = \sin(\pi t)/(\pi t)$, with $\operatorname{sinc}(0) = 1$. For the analytic construction in this section assume additionally that $\sqrt{f}$ is smooth in the interior of I. The numerical profiles are strictly positive finite cosine sums and satisfy this assumption. The variational problems later in the paper are defined on all of $L^2(I)$. Choose smooth even cutoffs $0 \le \chi_\varepsilon \le 1$, supported inside I, tending to one, and set

$$m_\varepsilon = \int f\chi_\varepsilon^2, \qquad f_\varepsilon = f\chi_\varepsilon^2/m_\varepsilon = \eta_\varepsilon^2.$$

Then $f_\varepsilon \to f$ in L^1 and L^2, and

$$K_\varepsilon(0) = 1, \quad |K_\varepsilon| \le 1, \quad \sup_{t\in\mathbb{R}} |K_\varepsilon(t) - K_f(t)| \le \|f_\varepsilon - f\|_1 \longrightarrow 0. \tag{2.2}$$

Let $N(T)$ count all nontrivial zeros ρ with $0 < \Im\rho \le T$, with analytic multiplicity. Write $L = \log T$ and $z_\rho = i(\rho - 1/2)L/(2\pi)$. Reflection $\rho \mapsto 1 - \bar\rho$ becomes $z \mapsto \bar z$. On the finite-dimensional span of

$F_z(u) = \eta_\varepsilon(u)e^{-2\pi iuz}$ consider

$$\mathcal{A} = \sum_z \nu_z F_z \langle F_{\bar{z}}, \cdot \rangle, \qquad \operatorname{tr}\mathcal{A} = N(T), \quad \operatorname{tr}\mathcal{A}^2 = \sum_{z,w} \nu_z \nu_w K_\varepsilon(z-w)^2. \tag{2.3}$$

The inner product is conjugate-linear in its first argument. Conjugate pairs make $\mathcal{A}$ Hermitian. The square in the last expression is not a modulus square; the whole sum is nevertheless the nonnegative trace of the square of a Hermitian operator.

The unconditional analytic input, in the form of [3, Lemma 3.1 and Section 3] and [10, Proposition 3.1], is

$$\operatorname{tr}\mathcal{A}^2 = (R(f_\varepsilon) + o(1))N(T), \qquad R(f_\varepsilon) \longrightarrow R(f). \tag{2.4}$$

The first limit keeps ε fixed. To specify the normalization, put $g = f_\varepsilon * f_\varepsilon$. Then

$$g - \widehat{g''/(4L^2)}(x) = (1 + \pi^2 x^2/L^2)K_\varepsilon(x)^2.$$

At $x = z_\rho - z_{\rho'}$ the factor cancels the weight $4/(4-(\rho-\rho')^2)$ in the unconditional pair formula [1]. Both g and g'' are fixed admissible smooth tests; the derivative contribution is $O_\varepsilon(N/L^2)$, while the first contributes $g(0) + \int |u| g(u)\,\mathrm{d}u = R(f_\varepsilon)$. The scaled real coordinates have total span at most $T\log T/(2\pi) = N(T) + o(N(T))$.

2.2. **The finite inequality and simple critical-line zeros.** For $c \geq 0$ let

$$\phi_c(x) = x^2 - (x-c)_+^2. \tag{2.5}$$

It is convex, quadratic below c, and linear above c. If $A = P + Q$ is Hermitian, $P \succeq 0$, and Q has at most p positive eigenvalues, then for $t \geq c \geq 0$,

$$\operatorname{tr} A^2 \geq \operatorname{tr}\phi_c(P) + 2t(\operatorname{tr} A - \operatorname{tr} P) - t^2 p. \tag{2.6}$$

Indeed write $Q = Q_+ - Z$, where $Z \succeq 0$ and $ZQ_+ = 0$. The cross term gives $\operatorname{tr} A^2 \geq \operatorname{tr}(P-Z)^2 + \operatorname{tr} Q_+^2$. For $X = P - cI$, the inequalities $\operatorname{tr}(XZ) \leq \operatorname{tr}(X_+ Z)$ and $\operatorname{tr}(X_+ - Z)^2 \geq 0$ imply $\operatorname{tr}(P-Z)^2 + 2c\operatorname{tr} Z \geq \operatorname{tr}\phi_c(P)$. Also $\operatorname{tr} Q_+^2 \geq 2t\operatorname{tr} Q_+ - t^2 p$. Adding leaves $2(t-c)\operatorname{tr} Z \geq 0$, proving (2.6). This is the finite clipping argument of [19, 10], recalled to fix every coefficient.

For any principal-block partition of a Hermitian matrix G,

$$\operatorname{tr}\phi_c(G) \geq \sum_B \operatorname{tr}\phi_c(G_B). \tag{2.7}$$

Choose an orthonormal eigenbasis of each block and apply scalar Jensen to the spectral measure of G at each resulting basis vector. Summing proves the assertion; operator convexity is not needed.

Let $n = N_{\mathrm{s}0}(T)$ be the number of simple critical-line zeros, and take P to be their part of (2.3). Its nonzero spectrum is that of the Gram matrix $U = (K_\varepsilon(y_i - y_j))$, and $\operatorname{tr} P = n$. Each remaining real location contributes at most one positive direction and has multiplicity at least two. Each reflected off-line pair of multiplicity j contributes $j(vw^* + wv^*)$, of trace $2j$ and with at most one positive direction, since

$$vw^* + wv^* = \tfrac{1}{2}\big((v+w)(v+w)^* - (v-w)(v-w)^*\big).$$

Thus $2p \leq N - n$. With $c = t = 2$,

$$\operatorname{tr}\mathcal{A}^2 \geq 2(N-n) + \operatorname{tr}\phi_2(U). \tag{2.8}$$

Suppose that $a < 1$, $b \geq 0$, and a constant B satisfy, for every finite real configuration of n points and all small ε,

$$\operatorname{tr}\phi_2(U) - n \geq an - bS - \omega(\varepsilon)n - B, \qquad \omega(\varepsilon) \longrightarrow 0, \tag{2.9}$$

where S is its span (zero for at most one point). Combining (2.4)–(2.9), and taking $T \to \infty$ before $\varepsilon \to 0$, gives

$$\liminf_{T\to\infty} \frac{N_{\mathrm{s}0}(T)}{N(T)} \geq \frac{2 - R(f) - b}{1-a}. \tag{2.10}$$

This is the standard certificate interface [2, 3, 4, 10]. The numerical application below keeps its profile, finite local input, and global Gram operator compatible throughout.

3. A profile for which separated energy is retained

The numerical construction uses the interval $I = [-1/2, 1/2]$ and the positive profile

$$v(s) = \cos(\sqrt{2}s) + \sum_{j=1}^{12} d_j \cos(2\pi j s), \qquad f(s) = \frac{v(s)}{K_0}, \qquad K_0 = \sqrt{2}\sin\theta, \quad \theta = \frac{\sqrt{2}}{2}. \tag{3.1}$$

The density f is extended by zero outside I. The coefficients are specified exactly by

j	$10^9 d_j$	j	$10^9 d_j$
1	12310798	7	−6846472
2	−15041681	8	3781297
3	3867664	9	−5670353
4	6489926	10	3355089
5	−992327	11	−450523
6	5580097	12	−218483

This is the cosine family introduced in [6], with the coefficients and eight-point inequality of [25]. We apply that inequality to the full clipped Gram energy. Close-pair matching, separated-set norm bounds, and convex pinching are established parts of this method [11, 10]; the estimates here realize their requirements for the specific profile (3.1). In fact, its coefficient absolute sum also lies within the uniform cosine-family range of [11, Section 3]. Thus the proportion obtained below is a new evaluation of the established transfer, rather than a new general separation theorem. The explicit estimates given here provide a self-contained realization at separation 17/20.

The normalization is immediate because every correction cosine has integral zero. Moreover,

$$\sum_{j=1}^{12} |d_j| \le \frac{13}{200}, \qquad \cos(\sqrt{2}s) \ge 1 - s^2,$$

so $v(s) \ge 137/200$ on I. In particular f is a probability density, and $|K_f(t)| \le 1$ for real t.

Its energy has a particularly simple exact expression. Set

$$P = \sum_{j=1}^{12} d_j^2 \left(\frac{1}{2} - \frac{1}{4\pi^2 j^2}\right), \qquad H = 2 - R(f) = \frac{3}{2} - \frac{\cos\theta}{\sqrt{2}\sin\theta} - \frac{P}{2\sin^2\theta}. \tag{3.2}$$

To see the cancellation behind this formula, put $v_0(s) = \cos(\sqrt{2}s)$. The function

$$v_0(s) + \int_I |s-t| v_0(t)\, dt$$

has zero second derivative and zero derivative at $s = 1/2$; it is therefore the constant $\cos\theta + K_0/2$. All cross terms with a mean-zero perturbation vanish in the energy form. For $j \ge 1$, direct differentiation gives

$$\int_I |s-t| \cos(2\pi j t)\, dt = \frac{(-1)^j - \cos(2\pi j s)}{2\pi^2 j^2}.$$

Orthogonality of the cosines then gives the correction P, while the unperturbed energy is $K_0\cos\theta + K_0^2/2$. Division by $K_0^2 = 2\sin^2\theta$ proves (3.2).

Here and below the numerical enclosures can be checked with rational arithmetic. One convenient enclosure is

$$H \ge H_0 := \frac{67216841}{10^8}. \tag{3.3}$$

For completeness, introduce the rational numbers

$$r_- = \frac{7071067811865475244}{10^{19}}, \qquad r_+ = \frac{7071067811865475245}{10^{19}},$$
$$s_- = \frac{64963693904}{10^{11}}, \qquad c_+ = \frac{76024459708}{10^{11}},$$
$$p_+ = \frac{314159265358979323847}{10^{20}}.$$

Squaring brackets θ between r_- and r_+. Taylor bounds give $\sin\theta \geq s_-$ and $\cos\theta \leq c_+$, and $\pi < p_+$. For example, the sine and cosine polynomials of degree at most eleven have error at most $13|x|^{12}/5748019200$ for $|x| \leq 1$; applying them at the rational endpoints proves the two trigonometric bounds. With

$$P_+ = \sum_{j=1}^{12} d_j^2 \left(\frac{1}{2} - \frac{1}{4p_+^2 j^2}\right),$$

the remaining check is the exact rational inequality

$$\frac{3}{2} - \frac{c_+}{2r_-s_-} - \frac{P_+}{2s_-^2} \geq \frac{67216841}{10^8}. \tag{3.4}$$

The positive denominators and $0 \leq P \leq P_+$ give the direction required in (3.3).

3.1. **A uniform norm bound from separation.** Let $d = 17/20$ and use the normalization $\operatorname{sinc}(t) = \sin(\pi t)/(\pi t)$. Consider

$$g(s) = \frac{7}{10}\left\{\operatorname{sinc}\big(d(s-1/2)\big) + \operatorname{sinc}\big(d(s+1/2)\big)\right\}^2. \tag{3.5}$$

This nonnegative integrable function satisfies

$$f(s) + \frac{5789}{960000} \leq g(s) \quad (s \in I), \qquad \Lambda := \int_{\mathbb{R}} g(s)\,ds \leq \frac{1253}{650} < 2, \qquad \widehat{g}(t) = 0 \quad (|t| \geq d). \tag{3.6}$$

The strict margin and the Fourier support, rather than pointwise convergence of matrix entries, are what make the resulting norm bound independent of the number of points.

The pointwise inequality admits a short polynomial verification. Since $K_0^{-1} \leq 109/100$, the elementary upper bound for cosine gives, with $x = s^2 \in [0, 1/4]$,

$$f(s) \leq \frac{109}{100}\left(\frac{213}{200} - x + \frac{x^2}{6}\right).$$

To bound the sum of sincs from below, write $D = \pi d$ and use

$$D^2 \leq \frac{357}{50}, \qquad D^4 \geq \frac{31684}{625}, \qquad D^6 \leq 364.$$

These follow from $157/50 \leq \pi \leq 22/7$. The degree-six lower bound for $\sin z/z$, applied at $D(s-1/2)$ and $D(s+1/2)$, gives the polynomial

$$2 - \frac{119}{200} + \frac{7921}{150000} - \frac{13}{5760} + \left(-\frac{119}{50} + \frac{7921}{6250} - \frac{13}{96}\right)x + \left(\frac{31684}{37500} - \frac{13}{24}\right)x^2 - \frac{13}{90}x^3.$$

Its constant exceeds $29/20$, its linear coefficient exceeds $-5/4$, and its last two terms have nonnegative sum on $[0, 1/4]$. Consequently the sinc sum is at least $29/20 - 5x/4 > 0$. Subtracting the displayed upper bound for f from the corresponding lower bound for g leaves

$$Q(x) = \frac{3109}{10000} - \frac{579}{400}x + \frac{2189}{2400}x^2.$$

Here

$$Q(x) - Q(1/4) = (1/4 - x)\left\{\frac{579}{400} - \frac{2189}{2400}(x + 1/4)\right\} \geq 0, \qquad Q(1/4) = \frac{5789}{960000},$$

which proves the required margin. The degree-seven lower Taylor bound for sine also gives

$$K_0 \geq 1 - \frac{1}{12} + \frac{1}{480} - \frac{1}{40320} > \frac{100}{109},$$

and hence the bound $K_0^{-1} \leq 109/100$ used above.

The Fourier transform of each sinc in (3.5) is a modulated interval indicator. The square corresponds to convolution, so its transform is supported on $[-d, d]$, is continuous, and vanishes at both endpoints. Parseval gives

$$\Lambda = \frac{28}{17}\{1 + \operatorname{sinc}(17/20)\}.$$

For a rational bound on the last quantity, put $y = 3\pi/20$ and use $\sin(17\pi/20) = \sin y$. The estimates $y^2 \geq 22206/10^5$ and $y^4 \leq 4932/10^5$ imply

$$\operatorname{sinc}(17/20) \leq \frac{3}{17}\left(1 - \frac{y^2}{6} + \frac{y^4}{120}\right) \leq \frac{3}{17}\frac{963401}{10^6}.$$

Substitution proves the bound $\Lambda \leq 1253/650$.

For any d-separated real points $x_1, \ldots, x_m$ and complex coefficients $z_1, \ldots, z_m$, (3.6) therefore gives

$$\begin{aligned} \sum_{i,j} \overline{z_i} z_j K_f(x_i - x_j) &= \int_I f(s) \left| \sum_j z_j e^{-2\pi i x_j s} \right|^2 ds \\ &\leq \int_{\mathbb{R}} g(s) \left| \sum_j z_j e^{-2\pi i x_j s} \right|^2 ds = \Lambda \sum_j |z_j|^2. \end{aligned} \tag{3.7}$$

The identity uses the vanishing of the off-diagonal Fourier terms, including pairs at distance exactly d. Thus the Gram matrix on every such set has spectrum below two.

The same conclusion applies after the smoothing used in the trace formula. Indeed $m_\varepsilon f_\varepsilon = f\chi_\varepsilon^2 \leq f$ and $m_\varepsilon \to 1$, so the smoothed Gram matrix has norm at most $\Lambda/m_\varepsilon < 2$ for all sufficiently small ε. This choice of ε works simultaneously for all separated sets and all their cardinalities.

3.2. **What a close pair contributes.** The complementary estimate is

$$K_f(t) \geq \frac{2583}{20000} > \frac{1}{8} \quad (|t| \leq 17/20). \tag{3.8}$$

Here is a direct bound that also keeps removable singularities out of the argument. Put

$$K_{\mathrm{MT}}(t) = \int_I \cos(\sqrt{2}s)\cos(2\pi t s)\, ds.$$

For $0 \leq t \leq d$, this function decreases with t, because the cosine factor is nonnegative and the other cosine decreases pointwise when expressed using $|s|$. At the endpoint, elementary integration gives, with $z = 17\pi/20$, $y = 3\pi/20$, and $\gamma = \cos\theta/K_0$,

$$\frac{K_{\mathrm{MT}}(d)}{K_0} = \frac{\cos y + 2\gamma z \sin y}{2z^2 - 1} \geq \frac{1}{5}.$$

For the last comparison it suffices to use

$$\gamma \geq \frac{4}{5}, \quad \frac{8}{3} \leq z \leq \frac{187}{70}, \quad \sin y \geq \frac{9}{20}, \quad \cos y \geq \frac{22}{25}.$$

The numerator is then at least $14/5$ and the positive denominator at most $32519/2450$. The sine and cosine bounds follow from their first Taylor inequalities and $471/1000 \le y \le 33/70$. For γ, use $\cos\theta \ge 3/4$ and $K_0 \le 1 - 1/12 + 1/480 = 147/160$, which give $\gamma \ge 40/49 > 4/5$.

The transform of each correction cosine is

$$\frac{1}{2}\{\operatorname{sinc}(t-j) + \operatorname{sinc}(t+j)\},$$

of absolute value at most one. The total correction to K_{MT}/K_0 is therefore at most

$$(109/100)(13/200) = 1417/20000.$$

Combining this with the preceding endpoint estimate proves (3.8); evenness covers negative t.

4. The eight-point floor and its global consequence

Take nonnegative gaps $g_0, \dots, g_6$, and put $p_0 = 0$, $p_j = \sum_{r<j} g_r$ for $1 \le j \le 7$. The prior eight-point certificate [25] is the inequality

$$\sum_{r=0}^{6} b_r g_r + \sum_{0 \le i < j \le 7} a_{ij} K_f(p_j - p_i)^2 \ge a_8 := \frac{805124}{10^8} \qquad (g_r \ge 0). \tag{4.1}$$

All entries in the following table are the numerators of $10^8 a_{ij}$; the lower triangle is unused.

$i\backslash j$	1	2	3	4	5	6	7
0	13630830	0	19565904	20317441	45030671	100000000	200000000
1		29997996	30621878	47766489	79682558	109938656	100000000
2			37356140	69378121	65335210	79682558	45030671
3				38030065	69378121	47766489	20317441
4					37356140	30621878	19565904
5						29997996	0
6							13630830

The pressure coefficients are

$$10^8(b_0, \dots, b_6) = (28898, 57272, 75526, 80958, 75526, 57272, 28898), \qquad B_8 := \sum_r b_r = \frac{404350}{10^8}. \tag{4.2}$$

The coefficients are nonnegative. Their seven lag sums are

$$10^8 \left(\sum_{i=0}^{7-\ell} a_{i,i+\ell} \right)_{\ell=1}^{7} = (199999997, 199999998, 199999996, \\ 199999998, 199999998, 200000000, 200000000), \tag{4.3}$$

so each lag has capacity at most two.

The local floor (4.1) is the computer-assisted part of this input. Its domain and every coefficient are specified here, while the accompanying formal source contains its complete finite proof data. Since every b_r is positive, a gap with $b_r g_r \ge a_8$ is handled by pressure alone. On the remaining bounded box the verification uses rational cells with retained pair-distance constraints, proved analytic minorants for K_f^2, and nonnegative multiplier combinations. Infeasible cells are excluded by rational linear combinations of their constraints; the other leaves verify the stated floor. The coefficient table is invariant under reversing the seven gaps, so a certified half-domain suffices. The analytic enclosures, covering argument, and exact leaf data are included in that source. Thus the numerical floor is a specified reusable mathematical input; the finite argument that follows only requires (4.1) and the visible capacities (4.3).

Consider first ordered points $x_0 < \cdots < x_{m-1}$ with $m \geq 8$. Sum (4.1) over all consecutive eight-point windows. If $S = x_{m-1} - x_0$, the pressure telescopes exactly to

$$\sum_{r=0}^{6} b_r(x_{m-7+r} - x_r) \leq B_8 S. \tag{4.4}$$

Each pair of index lag $\ell \leq 7$ is charged at most the full lag sum in (4.3). Hence

$$E_7(x) := 2 \sum_{\substack{0 \leq i < j < m \\ j-i \leq 7}} K_f(x_j - x_i)^2 \geq a_8(m-7) - B_8 S. \tag{4.5}$$

For $m \leq 7$ the right side is nonpositive, so this formula also holds for short configurations, with zero span for an empty configuration.

On a d-separated set the Gram matrix $U = (K_f(x_i - x_j))$ has norm below two by (3.7). Since its diagonal is one,

$$\operatorname{tr} \phi_2(U) - m = \operatorname{tr}(U - I)^2 = \sum_{i \neq j} |U_{ij}|^2 \geq a_8 m - B_8 S - 7a_8. \tag{4.6}$$

The entire off-diagonal energy is available on this block; selecting the seven-band energy in (4.5) only makes a lower bound smaller.

For arbitrary distinct real points, choose a maximal collection of disjoint pairs at distance less than d. If two unmatched points were closer than d, they could be added as another pair; thus all unmatched points form one separated remainder. A two-point Gram block has eigenvalues $1 \pm |K_f(t)|$, both in $[0, 2]$, and therefore contributes the clipped excess $2K_f(t)^2$. By (3.8), this is at least $1/32 > 2a_8$.

Convex spectral pinching permits these pair blocks and the single remainder block to be treated together. More explicitly, for any principal-block partition, choose an orthonormal eigenbasis in each block. Jensen's scalar inequality for the spectral measure of the full matrix, summed over the resulting orthonormal basis, gives $\operatorname{tr} \phi_2(U) \geq \sum_B \operatorname{tr} \phi_2(U_B)$. The remainder has span no larger than the original span, and it alone pays the pressure and endpoint term. We obtain

$$\operatorname{tr} \phi_2(U) - n \geq a_8 n - B_8 S - 7a_8 \tag{4.7}$$

for every finite configuration of n distinct real points.

To use this inequality with smooth trace tests, set $\delta_\varepsilon = \|f_\varepsilon - f\|_1$. Uniformly for real t, $|K_{f_\varepsilon}(t) - K_f(t)| \leq \delta_\varepsilon$, and both kernels have absolute value at most one. Each squared entry thus changes by at most $2\delta_\varepsilon$. The ordered seven-band has at most $14m$ entries, so its loss is at most $28\delta_\varepsilon m$. A two-point block loses at most $4\delta_\varepsilon$. Together with the uniform separated norm bound proved above, the same matching argument gives

$$\operatorname{tr} \phi_2(U_\varepsilon) - n \geq (a_8 - 28\delta_\varepsilon)n - B_8 S - 7a_8, \qquad \delta_\varepsilon \longrightarrow 0. \tag{4.8}$$

This error is proportional to the number of points. The proof never sums uniform approximation errors over all n^2 matrix entries.

4.1. **The zero-counting proportion.** Apply (4.8) to the simple critical-line part of the signed zero operator. The two-level trace inequality with clipping and comparison level both equal to two gives

$$\operatorname{tr} \mathcal{A}^2 \geq 2(N - n) + \operatorname{tr} \phi_2(U_\varepsilon).$$

The scaled span is at most $N + o(N)$. Keep the smoothing fixed while height tends to infinity, use the trace energy formula, and then let ε tend to zero. Since $a_8 < 1$, this yields

$$\liminf_{T \to \infty} \frac{N_{\mathrm{s}0}(T)}{N(T)} \geq \frac{H - B_8}{1 - a_8} \geq q := \frac{66812491}{99194876} = 0.673547805029767868\ldots. \tag{4.9}$$

The exact rational step is

$$(1-a_8)\frac{66812491}{99194876} = H_0 - B_8.$$

All displayed constants satisfy the needed guards $0 < a_8 < 1/64$, $B_8 \geq 0$, and $\Lambda < 2$. In particular the simple critical-line proportion is at least 67.35478050%, with that terminating percentage rounded down.

5. What the energy says about the different counts

Simplicity, critical-line location, and multiplicity give different ways to spend the same energy budget. We write $N_{\mathrm{s}0}(T)$ for simple critical-line zeros, $N_{\mathrm{s}}(T)$ for simple zeros anywhere, $N_0(T)$ for critical-line zeros with multiplicity, and $N_{\mathrm{d}}(T)$ for distinct complex zero locations. The count $N_{\mathrm{s}\cup 0}(T)$ includes the multiplicity mass of zeros that are simple or on the line, with the overlap counted once. Finally, $M_{\leq r}(T)$ is the mass at locations of multiplicity at most r, and $D_{\leq r}(T)$ is the number of those locations. Distinct locations with the same ordinate remain distinct. We suppress T inside the finite counting arguments.

The resulting lower limiting proportions are collected in Table 1. Each complete deduction uses its own profile and that profile's energy. The strengthened simple-block certificate supplies the first and fourth rows. The labelled local inequality supplies the second, and its elementary counting consequence supplies the third. The remaining rows use the third local profile with the multiplicity accounting developed in [3, 4, 10].

Table 1. Lower bounds for the liminf of each displayed ratio as $T \to \infty$. Percentages are rounded down to eight decimal places.

Ratio	Lower percentage	Input to the deduction
$N_{\mathrm{s}0}/N$	67.35478050	Strengthened simple-block floor
N_{d}/N	83.90046180	Labelled local floor
N_{s}/N	67.80092360	Distinct-count identity
$(N_{\mathrm{s}}+N_0)/(2N)$	83.67739025	Retained excess
$N_{\mathrm{s}\cup 0}/N$	88.93165002	Third-profile tangent
$M_{\leq 2}/N$	83.79775462	Third-profile mass costs
$D_{\leq 2}/N_{\mathrm{d}}$	93.94524765	Third-profile location costs

The strengthened certificate has

$$\widehat{a} = \frac{805124}{10^8}, \qquad \widehat{b} = \frac{404350}{10^8}, \qquad R_* \leq R_*^+ := 2 - \frac{67216841}{10^8}. \tag{5.1}$$

Consequently its simple-critical-line proportion is

$$q_* := \frac{2 - R_*^+ - \widehat{b}}{1-\widehat{a}} = \frac{66812491}{99194876}. \tag{5.2}$$

The coefficient $\widehat{a}$ is positive. This matters for the average consequence, because the finite argument preserves more information than its final numerical bound.

5.1. **The excess and the average count.** Let $n = N_{\mathrm{s}0}$, and let b_{exc} count one orbit for each multiple critical-line location and one for each reflected off-line pair. Its excess mass is

$$e = N - n - 2b_{\mathrm{exc}} \geq 0.$$

A multiple real location of multiplicity j contributes $j-2$ to e, while a reflected pair of multiplicity j at each member contributes $2(j-1)$. Before the positive-index count is replaced by $(N-n)/2$, the clipping inequality and a simple-block certificate with coefficients a, b give

$$\operatorname{tr}\mathcal{A}^2 + bS + o(N) + O(1) \geq 2N - (1-a)n + 2e. \tag{5.3}$$

Here the smoothing error is removed after the height limit, as in the preceding construction.

Let B_{off} be the mass of multiple off-line zeros. The contribution of a pair with multiplicity $j \ge 2$ is $2j$ to B_{off} and $2(j-1)$ to e, so $B_{\mathrm{off}} \le 2e$. Inclusion–exclusion for the two multiplicity counts is the exact identity

$$N_{\mathrm{s}} + N_0 = N + n - B_{\mathrm{off}}.$$

Combining it with (5.3) yields

$$N_{\mathrm{s}} + N_0 \ge 3N - \operatorname{tr} \mathcal{A}^2 - bS + an - o(N) - O(1).$$

For $a \ge 0$, insert the simple-critical lower bound $q = (2 - R^+ - b)/(1-a)$ and the trace and span limits. The identity $3 - R^+ - b + aq = 1 + q$ then gives

$$\liminf_{T\to\infty} \frac{N_{\mathrm{s}}(T) + N_0(T)}{2N(T)} \ge \frac{1+q}{2}. \tag{5.4}$$

At (5.2) the right side is $166007367/198389752$. This is a lower bound for one average; it does not separately bound either component by that number. The retained-excess deduction is part of the earlier counting framework [4, 10].

5.2. **Profile energies and the labelled distinct count.** For a finite cosine profile $f(u) = 1 + \sum_{j=1}^{D} a_j \cos(2\pi j u)$ on $[-1/2, 1/2]$, direct integration gives

$$R(f) = \frac{4}{3} + \frac{1}{2}\sum_{j=1}^{D} a_j^2 + \frac{1}{\pi^2}\sum_{j=1}^{D} \frac{(-1)^j a_j - a_j^2/4}{j^2}. \tag{5.5}$$

The first and third profiles of [10] have coefficient lists, in units of 10^{-6},

$$(193305, -23105, 29166, 13952, -21009, 41052), \qquad (200566, 19, -9358, 9283, -6746, 8170).$$

Rational enclosures in (5.5) give

$$R_1 \le R_1^+ := \frac{666352479141}{500000000000}, \qquad R_3 \le R_3^+ := \frac{83278977073}{62500000000}. \tag{5.6}$$

These evaluate the existing profiles more closely; their local inequalities are the cited computer-assisted inputs.

For the distinct count the first profile is used with

$$c = \frac{67}{20}, \qquad x_1 = \frac{54350}{10^6}, \quad x_2 = \frac{69433}{10^6}, \quad b_1 = \frac{37314}{10^6}.$$

If U is the Gram matrix, formed with the smoothed kernel, of real locations carrying labels $d_i \in \{1, 2\}$, set $G = D^{1/2} U D^{1/2}$, where $D = \operatorname{diag}(d_i)$. The labelled input is

$$\operatorname{tr} \phi_c(G) - \operatorname{tr} D^2 \ge x_1 n_1 + x_2 n_2 - b_1 S - \omega(\varepsilon)(n_1 + n_2) - O(1).$$

Here n_1, n_2 count simple and double real locations, and $\omega(\varepsilon) \to 0$; the height limit precedes removal of the smoothing. Retain all real locations in the positive part, pinch the higher multiplicities into scalar blocks, and leave reflected off-line pairs in the remainder. Write h for the number of higher-multiplicity real locations and k for the number of off-line pairs, so $N_{\mathrm{d}} = n_1 + n_2 + h + 2k$.

Put

$$\mu = 2 - 2x_1 + x_2, \qquad \nu = 2c - 3 + x_1 - x_2,$$
$$r_h = 6c - 7 - c^2 + x_1 - 2x_2, \quad r_k = 4c - 2 - c^2 - 2x_1.$$

Their values are

$$(\mu, \nu, r_h, r_k) = \left(\frac{1960733}{10^6}, \frac{3684917}{10^6}, \frac{224123}{125000}, \frac{43}{625}\right),$$

all positive. The energy debit is at most $(\nu+\mu)n_1 + (2\nu+\mu)n_2 + c^2(h+k)$. It is bounded by $\nu N + \mu N_{\mathrm{d}}$: a real location of multiplicity $j \ge 3$ leaves the nonnegative difference $j\nu + \mu - c^2 \ge r_h$,

and an off-line pair of multiplicity $j \geq 1$ leaves $2j\nu + 2\mu - c^2 \geq r_k$. The energy budget therefore gives

$$\liminf \frac{N_{\mathrm{d}}}{N} \geq \frac{3 - R_1^+ + x_2 - x_1 - b_1}{2 - 2x_1 + x_2} = \frac{822532020859}{980366500000}. \tag{5.7}$$

At each location, multiplicity one contributes one to $2N_{\mathrm{d}} - N$, multiplicity two contributes zero, and higher multiplicities contribute negatively. Thus $N_{\mathrm{s}} \geq 2N_{\mathrm{d}} - N$, and

$$\liminf \frac{N_{\mathrm{s}}}{N} \geq \frac{332348770859}{490183250000}. \tag{5.8}$$

This last count includes zeros off the critical line.

5.3. **The union and the full fixed-cutoff hierarchy.** For the remaining counts use the third profile's simple-block floor, with

$$a = \frac{54525}{10^6}, \qquad b = \frac{37272}{10^6}, \qquad \tau = R_3^+ + b = \frac{85608477073}{62500000000}, \qquad \beta = \tau - 1 - a = \frac{19700664573}{62500000000}. \tag{5.9}$$

Define

$$c_* = 2 + \sqrt{2 - 2a}, \qquad Z = \frac{c_*^2}{2} = 3 - a + 2\sqrt{2 - 2a}, \qquad E = \frac{\beta}{Z}. \tag{5.10}$$

In the finite clipping inequality use equal levels c_*. The simple-block floor remains valid because $\phi_c(s)$ increases with c for $s \geq 0$, and $c_* > 2$. In terms of $u = N - n$ and $e = u - 2b_{\mathrm{exc}}$, the resulting energy lower bound is

$$(1 + a)N + \left(2c_* - 1 - a - \frac{c_*^2}{2}\right) u + \frac{c_*^2}{2} e - bS - o(N) - O(1).$$

The coefficient of u vanishes exactly. Consequently

$$\limsup_{T \to \infty} \frac{e(T)}{N(T)} \leq E. \tag{5.11}$$

As $N - N_{\mathrm{s}\cup 0} \leq 2e$, this proves

$$\liminf \frac{N_{\mathrm{s}\cup 0}}{N} \geq 1 - 2E = 1 - \frac{2(\tau - 1 - a)}{3 - a + 2\sqrt{2 - 2a}}. \tag{5.12}$$

Equivalently, c_* is the largest permissible tangent level for which simple off-line pairs still cost at least $1 + a$ per unit mass; the cost of multiple off-line pairs increases up to that level.

The same excess controls every fixed higher cutoff. A real location of multiplicity $j \geq r + 1$ has mass-to-excess ratio $j/(j - 2) \leq (r + 1)/(r - 1)$. A nonreal pair has the smaller ratio $j/(j - 1)$. Hence, for every fixed integer $r \geq 3$,

$$\liminf \frac{M_{\leq r}}{N} \geq 1 - \frac{r + 1}{r - 1} E. \tag{5.13}$$

This is the retained multiplicity-budget mechanism of [26, 4], evaluated with (5.9). It uses the excess estimate, rather than the union bound alone.

For the corresponding location proportions, and for the stronger cutoff-two result, it is useful to give the common accounting once. Let x_j count real locations of multiplicity j, and y_j count reflected pairs of multiplicity j at each member. Then

$$N = \sum_j j x_j + 2 \sum_j j y_j, \qquad N_{\mathrm{d}} = \sum_j x_j + 2 \sum_j y_j.$$

Put $h_c(s) = 2cs - \phi_c(s)$. Retain all real locations in the positive part and pinch their Gram matrix into the simple block and scalar multiple-location blocks. For every $c \geq 2$, the finite inequality is

$$\operatorname{tr} \mathcal{A}^2 + bS + o(N) + O(1) \geq 2cN - \sum_j h_c(j) x_j - c^2 \sum_j y_j + a x_1. \tag{5.14}$$

All approximation errors can first be included in a fixed upper budget $\tau' N$, with $\tau' > \tau$. This also justifies the subsequent division by the distinct count.

Set $\alpha = 2c - \tau'$ and $\beta' = \tau' - 1 - a$. Moving the budget to the right gives

$$\sum_j \lambda_j x_j + \sum_j \omega_j y_j \le 0, \qquad \lambda_j = \alpha j - h_c(j) + a\mathbf{1}_{j=1}, \quad \omega_j = 2\alpha j - c^2.$$

If each low-multiplicity location costs at least $-\beta'$ and each high-multiplicity location at least $\gamma_r - \beta'$, then

$$0 \ge -\beta' N_{\mathrm{d}} + \gamma_r (N_{\mathrm{d}} - D_{\le r}), \qquad \frac{D_{\le r}}{N_{\mathrm{d}}} \ge 1 - \frac{\beta'}{\gamma_r}.$$

A reflected pair carries two location costs.

For $r \ge 3$, choose $c = c_* \in (3, 4)$. At the limiting budget, $\alpha = Z - \beta$ and $2c_* - 1 - c_*^2/2 - a = 0$. The real-location costs plus β at $j = 1, 2, 3$ are respectively 0, $3 - \tau - a$, and $8 - 2\tau - a$. For $j \ge 4$ the expression is $\alpha j + \beta - c_*^2$. The required guards are

$$\alpha > 0, \quad \beta > 0, \quad 3 - \tau - a > 0, \quad 8 - 2\tau - a > 0, \quad 4\alpha + \beta - c_*^2 = 2Z - 3\beta > 0.$$

They hold here: $a < 1/2$, $\tau < 7/5$, $0 < \beta < 1/3$, and $Z > 9/2$ suffice. The first high real location has cost plus β equal to

$$\gamma_r = (r+1)\alpha + \beta - c_*^2 = Z(r - 1 - rE) > 0.$$

For pairs, the lowest cost plus 2β is zero at $j = 1$, and the high-pair cost at $j = r + 1$ exceeds $2\gamma_r$ by c_*^2; all later costs increase. The strict guards persist for $\tau' > \tau$ sufficiently close to τ, while the zero guard for the simple pair is independent of τ'. Letting $\tau' \downarrow \tau$ gives

$$\liminf \frac{D_{\le r}}{N_{\mathrm{d}}} \ge 1 - \frac{E}{r - 1 - rE} \qquad (r \ge 3 \text{ fixed}). \tag{5.15}$$

These fixed-cutoff consequences are most useful at small cutoffs. For sufficiently large cutoffs, the unconditional exponential multiplicity estimates of [27] give stronger mass tails; the right side of (5.13) instead tends to $1 - E$.

5.4. **Multiplicity at most two and the sharp scalar population.** At $c = 3$ the real-location debits in (5.14) are $5 - a$, 8, and 9 for $j = 1$, $j = 2$, and $j \ge 3$. Every off-line pair has debit nine. Since $0 \le a \le 1/2$, the debit per unit mass at multiplicity at most two is at most $5 - a$; for the remaining mass it is at most three. Therefore

$$\liminf \frac{M_{\le 2}}{N} \ge \frac{3 - \tau}{2 - a} = \frac{101891522927}{121592187500}. \tag{5.16}$$

For location costs take $\alpha = 6 - \tau'$, $\beta' = \tau' - 1 - a$, and $\gamma_2 = 8 - 2\tau' - a$. The sufficient guards are

$$\beta' \ge 0, \quad \alpha > 0, \quad 3 - \tau' - a \ge 0, \quad 1 - 2a \ge 0, \quad \gamma_2 > 0.$$

The real costs satisfy $\lambda_1 = -\beta'$, $\lambda_2 + \beta' = 3 - \tau' - a \ge 0$, and $\lambda_j + \beta' - \gamma_2 = \alpha(j - 3) \ge 0$ for $j \ge 3$. For pairs, $\omega_1 + 2\beta' = 1 - 2a \ge 0$ and $\omega_3 + 2\beta' - 2\gamma_2 = 9$, after which the costs increase. All guards are strict at (5.9). The location argument followed by $\tau' \downarrow \tau$ gives

$$\liminf \frac{D_{\le 2}}{N_{\mathrm{d}}} \ge 1 - \frac{\tau - 1 - a}{8 - 2\tau - a} = \frac{101891522927}{108458411118}. \tag{5.17}$$

There is a direct reason that these two expressions fit together. Within the scalar orbit accounting of (5.14), they are attained simultaneously. Assign a fraction $\mu = (3 - \tau)/(2 - a)$ of the mass to simple real locations, and the remaining fraction to triple real locations. The energy charge is

$$(1 + a)\mu + 3(1 - \mu) = \tau.$$

The fraction of low-multiplicity locations is then

$$\frac{\mu}{\mu+(1-\mu)/3}=\frac{3\mu}{1+2\mu}=1-\frac{\tau-1-a}{8-2\tau-a}.$$

This population satisfies every scalar row obtained from (5.14) by changing $c\ge 2$: $\phi_c(1)=1$ and $\phi_c(3)\le 9$, with equality in the latter for $c\ge 3$. Its proportions are rational, so a common denominator produces integral location counts. Thus changing the clipping level alone cannot strengthen these two conclusions within that accounting. The population certifies sharpness of the reduced scalar constraints; it assigns no Gram interactions between its locations. Additional geometric or arithmetic information can impose further constraints.

The rational values in this section and the union radical in (5.12) determine all entries of Table 1. In particular, the larger simple-anywhere number, the average number, and the distinct-location number each retain their own numerator and denominator throughout the argument.

6. Local slack and continuous payment for long spans

6.1. **The fixed local input and its retained slack.** The main numerical bound is now established. We turn to a different question: what quantitative information is retained by local slack and by pair energy beyond the lags used in a local floor? An explicit application uses the second profile of [10, Section 5]:

$$\begin{aligned} f(u)&=1+\sum_{j=1}^{10}a_j\cos(2\pi ju),\\ 10^6(a_1,\dots,a_{10})&=(122290,-45012,16241,1887,2711,3731,\\ &\qquad -5307,2982,-5153,5358). \end{aligned}\tag{6.1}$$

The coefficient absolute sum is 0.210672, so $f>0$ and $|K_f|\le 1$. Throughout this section $K=K_f$, $\theta=17/20$, and

$$a=\frac{8722}{10^6},\quad \alpha=\frac{627}{10^6},\quad b=7\alpha=\frac{4389}{10^6},\quad d_0=\frac{10016}{10^6},\quad B_0=7a+\frac{3882}{10^6}.\tag{6.2}$$

The public eight-point certificate, with weights and bounded corrections listed in Appendix A, says that for every seven-gap frame with gaps at least θ,

$$F_i:=\alpha(y_{i+7}-y_i)+\sum_{0\le r<s\le 7}W_{rs}K(y_{i+s}-y_{i+r})^2+\sum_{r=0}^{6}B_r(g_{i+r})\ge a.\tag{6.3}$$

Its validity is the imported computer-assisted input [10, Proposition 5.3]. Define $\ell_i=F_i-a\ge 0$ and $H=\sum_i\ell_i$ over all complete frames. With ordered short-band energy

$$E_s=2\sum_{h=1}^{7}\sum_i K(y_{i+h}-y_i)^2,$$

the pair-capacity sums are at most two. The spread charge sums to at most $7\alpha S$; the bounded corrections telescope with loss at most $3882/10^6$. Retaining, rather than dropping, the slack gives

$$E_s\ge am-bS-B_0+H,\qquad \ell_i\ge\alpha(y_{i+7}-y_i)-d_0.\tag{6.4}$$

The second inequality uses nonnegative pair weights and $\sum_r\min B_r=-1294/10^6$. For fewer than eight points take $H=0$; the first inequality still holds because $B_0\ge 7a$.

6.2. **A general phase comparison.** The mechanism is more general than these data. Let K be real-valued on the real axis. Suppose frames of h gaps have nonnegative slack satisfying $\ell_i \ge \alpha(y_{i+h} - y_i) - d_0$, with $\alpha > 0$. Assume gaps are at least $\theta > 0$ and

$$|K(t)| \ge \frac{h_0 |\sin \pi t|}{\pi t} \quad (t \ge t_0), \qquad h_0 > 0. \tag{6.5}$$

Choose finitely many integer lags $l \ge h$ with $l\theta \ge t_0$ and nonnegative weights w_l satisfying

$$|e^{ix} - 1|^2 \le \sum_l w_l |e^{ilx} - 1|^2 \quad (x \in \mathbb{R}). \tag{6.6}$$

Put $q_l = \lceil l/h \rceil$ and choose positive numbers

$$R_l \ge q_l d_0/\alpha - (hq_l - l)\theta. \tag{6.7}$$

Writing $T = 2\sum_l \sum_i K(y_{i+l} - y_i)^2$, one obtains

$$\sum_i \operatorname{dist}(g_i, \mathbb{Z})^2 \le M_E T + M_H H + B_1, \tag{6.8}$$

where

$$M_E = \max_l \frac{w_l \pi^2 R_l^2}{8h_0^2}, \qquad M_H = \sum_l \frac{w_l q_l}{2\alpha R_l}, \qquad B_1 = \frac{1}{8}\sum_l w_l l. \tag{6.9}$$

Here and below empty sums are zero.

To prove this, set $c(t) = |e^{2\pi i t} - 1|^2 \le 4$. For every $R > 0$ and every $t \ge t_0$, (6.5) gives

$$c(t) \le \frac{4\pi^2 R^2}{h_0^2} K(t)^2 + \frac{8}{R}(t - R)_+. \tag{6.10}$$

For $t \le R$ the first term suffices. For $t > R$ it pays at least $c(t)R^2/t^2$; the remainder is at most $4(1 - R^2/t^2) \le 8(t - R)/R$.

A lag-l edge is covered by the $q = q_l$ complete frame starts $0, h, \dots, (q-2)h, l - h$ relative to its anchor, with a single frame when $l = h$. There are exactly $hq - l$ repeated gap incidences. Since each gap is at least θ, the sum L_i of the cover slacks satisfies

$$L_i \ge 0, \qquad L_i \ge \alpha t + \alpha(hq - l)\theta - q d_0 \ge \alpha(t - R_l).$$

Each fixed frame occurs in at most q_l translated covers for that lag. Summing (6.10) over its anchors consequently gives

$$\sum_i c(y_{i+l} - y_i) \le \frac{4\pi^2 R_l^2}{h_0^2} \sum_i K(y_{i+l} - y_i)^2 + \frac{8q_l}{\alpha R_l} H. \tag{6.11}$$

All frames lie inside their edge, and every physical span is included.

Now extend $z_i = e^{2\pi i y_i}$ by zero outside the finite configuration. Parseval applies (6.6) to its discrete differences. For lag l the zero extension adds $2\min(l, m) \le 2l$ unit boundary terms. Dropping the favorable nearest-neighbor boundary gives

$$\sum_i |z_{i+1} - z_i|^2 \le \sum_l w_l \sum_i |z_{i+l} - z_i|^2 + 2\sum_l w_l l.$$

Finally $\operatorname{dist}(g, \mathbb{Z})^2 \le |e^{2\pi i g} - 1|^2/16$. Insert (6.11) and retain the factor two in the ordered energy T to obtain (6.8).

There is a simple uniform family of phase comparisons. On $[r, 2r]$ take endpoint weights $2/r$ and interior weights $4/r$. For the unitary shift V, $V - I = (V^{l+1} - I) - V(V^l - I)$; the squared triangle inequality averaged over $r \le l < 2r$ proves (6.6). The weight sum is four and its first moment is $6r$. If $d_0 > 0$, the choice $R_l = q_l d_0/\alpha$ gives

$$M_H = 2/d_0, \qquad M_E \le \frac{\pi^2 d_0^2 \lceil 2r/h \rceil^2}{2h_0^2 \alpha^2 r}, \qquad B_1 = 3r/4. \tag{6.12}$$

Thus the energy price grows at most linearly in r, while the slack price is independent of r. If the local floor spends every lag through h, take $r > h$ before adding T to that floor.

For degree-D cosine profiles with $\sum |a_j| \le A_0$ and $f(1/2) \ge v_0 > 0$, the rational sinc factor satisfies

$$V(t^2) = f(1/2) + \sum_{j=1}^{D} (-1)^j a_j \frac{j^2}{t^2 - j^2}, \qquad |V(t^2) - f(1/2)| \le 2A_0 D^2/t^2 \quad (t^2 \ge 2D^2).$$

Hence $h_0 = v_0/2$ is available beyond $D\max(\sqrt{2}, 2\sqrt{A_0/v_0})$. More precisely, if $h, \theta, \alpha, d_0, A_0, v_0$ are fixed, with $\theta, \alpha, d_0, v_0 > 0$, a family satisfying the stated frame hypothesis permits $r > h$ with $r\theta$ beyond that threshold and $r = O(D)$ for $D \ge 1$. Formula (6.12) then gives an $O(D)$ energy price and a constant slack price. Each profile and its finite lag set are fixed before the height limit.

6.3. **An exact finite instance.** For (6.1), write $c_j = (-1)^j a_j$ and $z_0 = (51/5)^2$. Its sinc factor is

$$K(t) = \frac{\sin \pi t}{\pi t} V(t^2), \qquad V(z) = 1 + \sum_{j=1}^{10} c_j \frac{z}{z - j^2}.$$

For negative c_j use the value at z_0; for positive c_j use the limiting value c_j. This proves, for $t \ge 51/5$,

$$V(t^2) \ge \frac{3123731672835001}{3750959784000000} > \frac{104}{125} =: h_0. \tag{6.13}$$

Take

$$\mathcal{L} = \{12, \dots, 49\}, \quad q_l = \lceil l/7 \rceil, \quad R_l = \frac{201 q_l + 17 l}{20}, \quad w_l = \frac{196}{R_l^2}. \tag{6.14}$$

The identity $R_l = 16q_l - (7q_l - l)\theta$ and $16 > d_0/\alpha = 10016/627$ verify (6.7). The phase comparison is certified by the polynomial

$$P(t) = \frac{\sum_{l=12}^{49} w_l (1 - T_l(t)) - (1 - t)}{1 - t}, \tag{6.15}$$

where T_l is the Chebyshev polynomial. Its numerator vanishes at one, so P has degree 48. All 49 Bernstein coefficients on each of the following intervals exceed the displayed rational bound: The nonnegative Bernstein basis proves $P \ge 0$ on $[-1, 1]$ and therefore (6.6) on the whole real line.

TABLE 2. Certified lower bounds for every Bernstein coefficient of the phase polynomial on six intervals.

Interval	Lower bound
$[-1, -3/4]$	164977/50000000
$[-3/4, -1/2]$	255867387/1000000000
$[-1/2, 0]$	370473121/1000000000
$[0, 1/2]$	207778923/200000000
$[1/2, 3/4]$	743826529/250000000
$[3/4, 1]$	1129786507/125000000

Appendix B specifies the exact basis conversion. The prices in (6.9), using $\pi < 22/7$, satisfy

$$M_E < \frac{1890625}{5408} < 350, \qquad M_H < 112, \qquad B_1 < 6. \tag{6.16}$$

With $J = H + T$, (6.8) gives (1.2). The unused lags 12–49 are disjoint from the paid lags 1–7, so $D = E_s + T$ also satisfies

$$D \ge am - bS - B_0 + J. \tag{6.17}$$

6.4. **A signed pressure row for the retained phase resource.** For every $g \geq 17/20$ we prove

$$2K(g)^2 + pg + C\,\mathrm{dist}(g,\mathbb{Z})^2 \geq c, \qquad p = \frac{17}{4000}, \quad C = \frac{19}{4}, \quad c = \frac{1891}{200000}. \tag{6.18}$$

The profile bound $f \leq 1.210672$ gives $|K''| \leq \pi^2(1.210672)/3 < 4$. The sinc expansion

$$K(t) = \operatorname{sinc} t + \tfrac{1}{2}\sum_{j=1}^{10} a_j\{\operatorname{sinc}(t-j) + \operatorname{sinc}(t+j)\}, \qquad \operatorname{sinc} t = \frac{\sin \pi t}{\pi t},$$

gives the exact values

k	$K(k)$	$K'(k)$
1	12229/200000	−260321359/264000000
2	−11253/500000	2577793683/6160000000

Write $A_k = |K(k)|$, $B_k = |K'(k)|$ and take $r_1 = 1/20$, $r_2 = 1/25$. At both integers the signed derivative of $|K|$ is negative. Taylor's formula gives, for $d \geq 0$,

$$\operatorname{sign}(K(k))K(k+d) \geq A_k - B_k d - 2d^2, \qquad \operatorname{sign}(K(k))K(k-d) \geq A_k + B_k d - 2d^2.$$

On $0 \leq d \leq r_k$ the first right side is positive. For $g = k + d$ the left side of (6.18) minus c is at least

$$q_0 + q_1 d + q_2 d^2 + 8B_k d^3 + 8d^4, \quad q_0 = 2A_k^2 + pk - c, \quad q_1 = -4A_k B_k + p, \quad q_2 = 2B_k^2 - 8A_k + C.$$

Here $q_2 > 0$ and the exact positive margins $4q_0q_2 - q_1^2$ are

$$\frac{53039685134812231}{193600000000000000000}, \qquad \frac{239075721629797712 41}{948640000000000000000000} \tag{6.19}$$

for $k = 1, 2$. Completing the quadratic square proves the right-hand inner intervals. On the left, $|K(k-d)| \geq A_k + (B_k - 2r_k)d$ and

$$2K(k-d)^2 + p(k-d) + Cd^2 - c \geq q_0 + \{4A_k(B_k - 2r_k) - p\}d.$$

The two linear coefficients are respectively

$$\frac{2804524299211}{13200000000000}, \qquad \frac{1835448537709}{70000000000000},$$

both positive.

For $r_k \leq d \leq 1/2$, discard the kernel term. The lower expression $p(k-d) + Cd^2 - c$ increases with d and at the two cutoffs equals 2583/400000 and 259/40000. If a nearest integer is at least three, then $g \geq 5/2$ and $pg > c$. These cases cover every permitted gap, including half-integer ties, and prove (6.18) without sampled-grid acceptance.

6.5. **Cancellation of a single retained resource.** The following algebra explains the role of phase rigidity. Suppose $D \geq am - bS - B_0 + J$, $J \geq 0$, and $\sum \mathrm{dist}(g_i,\mathbb{Z})^2 \leq MJ + B_1$, while $D \geq 2\sum_i K(g_i)^2$. Summing a row of the form (6.18) gives the alternative bound

$$D \geq cm - pS - c - CB_1 - MCJ.$$

Taking convex weights $MC/(1+MC)$ and $1/(1+MC)$ cancels J. The resulting coefficients are

$$a' = a + \frac{c-a}{1+MC}, \qquad b' = b + \frac{p-b}{1+MC}, \qquad B' = \frac{MCB_0 + c + CB_1}{1+MC}. \tag{6.20}$$

This combines alternative bounds on the same energy; it does not add overlapping pair resources.

For $M = 350$, $B_1 = 6$ and (6.18), $1 + MC = 3327/2$ and

$$D \geq a_\star m - b_\star S - B_\star, \qquad B_\star = \frac{27293111}{332700000}, \tag{6.21}$$

where

$$a_\star = \frac{725489}{83175000}, \qquad b_\star = \frac{584077}{133080000}.$$

There are 45 retained index bands. Under $\sup|K_\varepsilon - K| \le \eta \le 1$, $|K_\varepsilon^2 - K^2| \le 3\eta$. Keep J defined on the original kernel. The first floor loses at most $270\eta m$; the adjacent-gap row loses at most $6\eta m$. Their same convex combination therefore gives

$$D(K_\varepsilon) \ge \left(a_\star - \frac{299254}{1109}\eta\right) m - b_\star S - B_\star. \tag{6.22}$$

The imported majorant for this profile has $\Lambda = 1.78942\ldots < 2$, and the near-pair bound is $\beta = 13707941/2^{28} > a_\star$ [10, Section 5]. Choose $\eta < (\beta - a_\star)/3$. Each removed pair pays more than $2a_\star$, the separated block is unclipped, and $b_\star > 0$. The matching argument of Section 4 transfers (6.22) to every real configuration. The coefficients satisfy $a_\star > a$ and $0 < b_\star < b$: the same local input has acquired a larger point reward and a smaller spread cost. This is a quantitative use of the retained phase resource, for the profile (6.1); it is not an additional gain in (1.1), which uses a different profile.

7. A common residual for joint affine deductions

The phase comparison retains local slack and unused pair energy on the same configuration. We now ask what can be deduced when the retained information has already been reduced to universal affine floors. The distinction is between precisely specified sets of inequalities, not between different assertions about the zeros themselves.

7.1. **The relaxation being tested.** Continue to write $I = [-1/2, 1/2]$. For a real function $f \in L^2(I)$, extended by zero, put

$$\begin{aligned} A(f) &= \int_I f(s)^2 \,\mathrm{d}s, \\ J(f) &= \iint_{I^2} |s-t| f(s) f(t) \,\mathrm{d}s\,\mathrm{d}t, \\ C_h(f) &= \int_{\mathbb{R}} f(s) f(s+h) \,\mathrm{d}s, \qquad R(f) = A(f) + J(f). \end{aligned} \tag{7.1}$$

The variational arguments will allow signed real functions with $\int f = 1$. A profile used in a Gram matrix is nonnegative; an analytic profile is also even and is subject to the fixed-test approximation in Section 2. Bounds proved on this larger space apply in particular to the nonnegative subclass.

For a normalized nonnegative profile let $G_f = (K_f(x_j - x_k))_{j,k=1}^n$ on a real configuration. It is a positive semidefinite Hermitian matrix with unit diagonal, including when f is not even. For $t \ge 0$ define

$$D_t(G_f) = \operatorname{tr}\big((G_f - I)^2 - (G_f - (1+t)I)_+^2\big). \tag{7.2}$$

The scalar function in this definition is nonnegative on $[0, \infty)$, and the subtracted square is nonnegative. Therefore

$$0 \le D_t(G_f) \le \operatorname{tr}(G_f - I)^2 = \sum_{j \ne k} |K_f(x_j - x_k)|^2. \tag{7.3}$$

Since $\operatorname{tr} G_f = n$, we also have $D_t(G_f) = \operatorname{tr} \phi_{1+t}(G_f) - n$.

Let b_{exc} count the exceptional conjugacy orbits: one for each multiple real location and one for each nonreal pair. The orbit argument in Section 2 gives $n_+(Q) \le b_{\mathrm{exc}}$ when the simple-real part is removed. In (2.6) take the two levels equal to $\Gamma/2$, where $\Gamma \ge 2$. It follows that

$$\operatorname{tr} \mathcal{A}_f^2 \ge \Gamma N - (\Gamma - 1)n - \frac{\Gamma^2}{4} b_{\mathrm{exc}} + D_{\Gamma/2-1}(G_f). \tag{7.4}$$

This is the compensated inertia inequality of [4]. It uses the same original multiset in every profile realization. The exceptional-orbit count is common to the inequalities; no rank statement for a sum of different profile operators is assumed.

Write the normalized omitted mass and excess as

$$u = \frac{N-n}{N}, \qquad e = \frac{N-n-2b_{\mathrm{exc}}}{N}, \qquad 0 \le e \le u \le 1. \tag{7.5}$$

When passing to limiting rows, take a common convergent subsequence of these bounded ratios and use the same symbols for its limit. This preserves one count point while the fixed-test error terms vanish. Nonnegativity of e follows orbit by orbit: a multiple real location of multiplicity k contributes $k-2$, and a nonreal pair with multiplicity j at each member contributes $2(j-1)$.

A row consists of finitely many fixed profiles f_i, weights $w_i \ge 0$ with $\sum_i w_i = 1$, and fixed thresholds $\Gamma_i \ge 2$. Its finite hypothesis is a genuinely joint floor

$$\sum_i w_i D_{\Gamma_i/2-1}(G_{f_i}) \ge an - dS - B, \qquad d \ge 0, \tag{7.6}$$

valid on every real configuration of size n and span S, with B independent of n. The same points occur in every matrix. No decomposition into separately minimized floors is required. All parameters, including the finite family and endpoint constant, are fixed before the height limit.

Average (7.4) with these weights. The trace formula and span bound of Section 2 give the normalized necessary row

$$\overline{R} + d \ge (1-u)(1+a) + \overline{\Gamma}u - \frac{u-e}{8}\sum_i w_i\Gamma_i^2, \quad \overline{R} = \sum_i w_i R(f_i), \quad \overline{\Gamma} = \sum_i w_i\Gamma_i. \tag{7.7}$$

The sign $d \ge 0$ is needed when replacing S/N by its upper limit one. For this analytic implication each profile is a fixed admissible smooth test, or a limit of such tests for which the finite-floor error is uniform in the configuration and any required norm control has been proved separately, as in Section 2. Continuity of matrix entries and spectral positive parts passes such a common finite floor to its L^2 limit on each fixed configuration. It does not transfer an arbitrary unsmoothed floor back to its approximants uniformly in the number of points. The periodic and variational arguments below are valid for all L^2 profiles without this additional analytic admissibility issue.

The relaxation studied here retains any collection of rows (7.7), their common count variables, and the orbit accounting for the desired statistic. Each row may have its own profiles, weights, slope and thresholds. It does not automatically retain labelled-block inequalities, square-completion slack, signed identities between exact energies, products of operators, or further arithmetic information. The conclusions concern precisely this relaxation. They impose neither an upper bound on an actual zero proportion nor a ceiling on every deduction from pair correlation. In particular they do not apply to the labelled distinct-zero scheme merely by changing the numerator.

7.2. **Periodic tests without a regularity assumption.** Take p distinct points $x_1, \dots, x_p$ modulo a period $L > 0$. Their full periodic energy per point, including the diagonal, is

$$Q_{L,x}(f) = \frac{1}{p}\sum_{a,b=1}^{p}\sum_{k\in\mathbb{Z}} |K_f(x_b - x_a + kL)|^2. \tag{7.8}$$

The sampling identity is valid for every real L^2 profile. With the Fourier convention (2.1), periodize $f(s)e^{2\pi ids}$ on an interval of length $1/L$:

$$P_d(s) = \sum_{\nu\in\mathbb{Z}} f(s+\nu/L)e^{2\pi id(s+\nu/L)}, \qquad 0 \le s < 1/L.$$

Only finitely many translates overlap, so $P_d \in L^2$. Its normalized Fourier coefficient of index k is $LK_f(d-kL)$. Parseval therefore gives

$$\sum_k |K_f(d+kL)|^2 = \frac{1}{L}\int_0^{1/L} |P_d(s)|^2\,\mathrm{d}s.$$

Expand the square, group by the difference of the translate indices, and sum over $d = x_b - x_a$. This proves

$$Q_{L,x}(f) = \frac{1}{pL} \sum_{j\in\mathbb{Z}} \left| \sum_{a=1}^{p} e^{2\pi i j x_a/L} \right|^2 C_{j/L}(f). \tag{7.9}$$

This is a finite sum, since $C_h(f) = 0$ for $|h| \geq 1$. There is no endpoint term at $|h| = 1$, where the supports intersect only on a null set. For an integer period it becomes

$$Q_{L,x}(f) = \frac{1}{pL} \left\{ p^2 A(f) + 2 \sum_{1\leq j<L} \left| \sum_{a=1}^{p} e^{2\pi i j x_a/L} \right|^2 C_{j/L}(f) \right\}. \tag{7.10}$$

This uses Hilbert-space Parseval rather than a pointwise Poisson-summation or Fourier-decay assertion.

Apply (7.6) to M consecutive periods. Their span divided by their number of points tends to L/p. Their off-diagonal square energy per point tends to $Q_{L,x}(f) - 1$: each fixed displacement occurs with a weight tending to one and bounded by one, while Parseval supplies the summable majorant. Consequently (7.3) gives

$$a \leq \sum_i w_i \{Q_{L,x}(f_i) - 1\} + dL/p. \tag{7.11}$$

We may average finitely many such inequalities with nonnegative weights. This does not require concatenating the periodic states or realizing their mixture as a single zero process.

7.3. **The residual identity and its quantifiers.** Fix $0 < u < 1$, put $q = 1 - u$, and choose a finite mixture of periodic states whose average gap is $1/q$. Let $Q_q(f)$ be its mixture of full per-point energies. For a row define

$$\mathcal{F}_q(f) = R(f) - qQ_q(f), \qquad \sigma = \sum_i w_i \{Q_q(f_i) - 1\} + d/q - a. \tag{7.12}$$

Equation (7.11) gives $\sigma \geq 0$. Let Res be the left side of (7.7) minus its right side. Substitution gives the exact identity, valid also at $e = u$,

$$\mathrm{Res} = \sum_i w_i \left\{ \mathcal{F}_q(f_i) - \Gamma_i u + \frac{\Gamma_i^2}{8}(u - e) \right\} + q\sigma. \tag{7.13}$$

For $0 \leq e < u$, completion of each scalar square gives

$$\begin{aligned} \mathrm{Res} = \sum_i w_i \mathcal{F}_q(f_i) - \frac{2u^2}{u-e} \\ + \frac{u-e}{8} \sum_i w_i \left(\Gamma_i - \frac{4u}{u-e} \right)^2 + q\sigma. \end{aligned} \tag{7.14}$$

The optimizing threshold $4u/(u-e) \geq 4$ is permitted. At $e = 0$ its contribution is $(u/8)\sum_i w_i(\Gamma_i - 4)^2$.

Suppose $\mathcal{F}_q$ has a minimizer $f_q^\dagger$ on the real affine space $\int f = 1$, with minimum m_q, and

$$\mathcal{F}_q(h) \geq \kappa_q \|h\|_2^2 \quad \left(\int h = 0 \right), \qquad \kappa_q > 0.$$

Stationarity makes the linear cross term vanish:

$$\mathcal{F}_q(f) = m_q + \mathcal{F}_q(f - f_q^\dagger). \tag{7.15}$$

Thus every row is satisfied at the same count point whenever

$$m_q \geq \frac{2u^2}{u-e}. \tag{7.16}$$

The residual retains exactly the profile departure, threshold departure and periodic-test slack; its profile contribution is at least $\kappa_q \sum_i w_i \|f_i - f_q^\dagger\|_2^2$.

The common point makes the obstruction simultaneous. The mixture and (u, e) are fixed before a row is chosen; only the nonnegative slack σ depends on that row. Arbitrarily many already valid limiting rows can therefore be retained together. This does not establish height-uniform control for a growing family of tests or growing thresholds.

No division by $u - e$ is made at $e = u$: (7.13) remains the appropriate formula. A point with $e = u > 0$ cannot survive the complete collection of all-threshold rows. Indeed fix one admissible profile and use $D_{\Gamma/2-1} \geq 0$, so $a = d = B = 0$. Its row requires $R(f) \geq 1 - u + \Gamma u$ for every fixed $\Gamma \geq 2$, which is impossible as Γ increases. The endpoint $u = e = 0$ is tested directly in (7.7) and is not represented by the quotient in (7.16).

8. An explicit two-lattice variational frontier

The simplest density-balanced mixture gives an exact two-count model and a quantitative rigidity identity. Its simple-critical endpoint is less restrictive numerically than the construction of Section 9, but the whole frontier has a closed form and applies to the union count.

8.1. **The quadratic form and its coercivity.** The unit lattice and the lattice of spacing two have full energies $A(f)$ and $A(f)/2 + C_{1/2}(f)$, respectively. For $1/8 \leq u \leq 1/3$, put $q = 1 - u$ and give the unit lattice weight $\theta = 2 - 1/q$. Then $\theta + 2(1 - \theta) = 1/q$, and the quadratic form is

$$F_u(f) = J(f) + \frac{3u}{2} A(f) - u C_{1/2}(f). \tag{8.1}$$

Its unique normalized real minimizer will be positive and even, with minimum

$$m(u) = \frac{1}{2} + u\{\alpha \cot \alpha - 2\beta \tan \beta\}, \qquad \alpha = \frac{1}{2\sqrt{2u}}, \quad \beta = \frac{1}{4\sqrt{u}}. \tag{8.2}$$

The estimate needed down to $u = 1/8$ is

$$F_u(h) \geq \left(u - \frac{1}{8} + \frac{1}{1000}\right) \|h\|_2^2 \qquad \left(\int_I h = 0\right). \tag{8.3}$$

To prove it put $H(s) = \int_{-1/2}^{s} h(t)\,\mathrm{d}t$. Both endpoint values of H vanish. For $g_h(s) = \int |s-t| h(t)\,\mathrm{d}t$, one has $g_h' = 2H$; integration by parts gives

$$J(h) = -2 \int_I H(s)^2\,\mathrm{d}s. \tag{8.4}$$

Let B be half the measure-preserving half-interval swap: $(Bh)(s) = h(s + 1/2)/2$ for $s < 0$ and $(Bh)(s) = h(s - 1/2)/2$ for $s > 0$. Then $B = B^*$, $B^2 = I/4$, and $C_{1/2}(h) = \langle h, Bh \rangle$. The forms $A, J, C_{1/2}$ preserve reflection parity, so their even–odd cross terms vanish.

If h is even and zero-mean, its primitive also vanishes at zero. Poincare's inequality on each half-interval gives $J(h) \geq -\|h\|_2^2/(2\pi^2)$. Since $C_{1/2}(h) \leq A(h)/2$,

$$F_{1/8}(h) \geq \left(\frac{1}{8} - \frac{1}{2\pi^2}\right) \|h\|_2^2 > \frac{1}{1000} \|h\|_2^2.$$

For odd h, write $h = t\psi + v$ with $\psi(s) = \sqrt{2}\sin(\pi s)$ and $v \perp \psi$ odd. The odd Neumann basis consists of $\sqrt{2}\sin((2j+1)\pi s)$, $j \geq 0$. Its primitives are orthogonal Dirichlet modes of squared norm $1/((2j+1)^2\pi^2)$. Hence

$$J(h) \geq -\frac{2}{\pi^2} t^2 - \frac{2}{9\pi^2} \|v\|_2^2.$$

Direct integration and $B^2 = I/4$ give

$$\langle \psi, B\psi \rangle = -1/\pi, \qquad \|B\psi + \psi/\pi\|_2^2 = 1/4 - 1/\pi^2.$$

Using $\langle v, Bv\rangle \le \|v\|_2^2/2$ now yields

$$F_{1/8}(h) \ge a_0 t^2 + b_0\|v\|_2^2 - 2c_0|t|\|v\|_2,$$

where

$$a_0 = \frac{3}{16} + \frac{1}{8\pi} - \frac{2}{\pi^2}, \quad b_0 = \frac{1}{8} - \frac{2}{9\pi^2}, \quad c_0^2 = \frac{1}{64}\left(\frac{1}{4} - \frac{1}{\pi^2}\right).$$

For a rational positivity check put $\pi_L = 157/50$, $\pi_U = 22/7$,

$$a_L = \frac{3}{16} + \frac{1}{8\pi_U} - \frac{2}{\pi_L^2}, \quad b_L = \frac{1}{8} - \frac{2}{9\pi_L^2}, \quad c_U^2 = \frac{1}{64}\left(\frac{1}{4} - \frac{1}{\pi_U^2}\right).$$

Both $a_L - 1/1000$ and $b_L - 1/1000$ are positive, and

$$(a_L - 1/1000)(b_L - 1/1000) - c_U^2 = \frac{17310471944257}{330823607944500000} > 0.$$

The two-by-two form is therefore at least $(t^2 + \|v\|_2^2)/1000$. This proves the estimate at $u = 1/8$. Finally

$$F_u(h) - F_{1/8}(h) = (u - 1/8)\{3A(h)/2 - C_{1/2}(h)\} \ge (u - 1/8)A(h),$$

which proves (8.3).

8.2. **The minimizer and its exact defect.** With α, β as in (8.2), set $p = \alpha/\sin\alpha$ and $z = -\beta/\cos\beta$, and define

$$f_u(s) = p\cos(\sqrt{2/u}(|s| - 1/4)) + z\sin((|s| - 1/4)/\sqrt{u}). \tag{8.5}$$

The sine term integrates to zero on each half-interval, and $4p\sin\alpha/\sqrt{2/u} = 1$, so $\int f_u = 1$. For $g(s) = \int |s - t| f_u(t)\,\mathrm{d}t$ the Euler expression on $0 < s < 1/2$ is

$$L(s) = 3uf_u(s) - uf_u(1/2 - s) + 2g(s).$$

The cosine component is symmetric about $1/4$ and solves $2uw'' + 4w = 0$; the sine component is antisymmetric and solves $4uw'' + 4w = 0$. As $g'' = 2f_u$, these imply $L'' = 0$. The endpoint derivatives are $f_u'(0+) = 1/(4u)$ and $f_u'(1/2-) = -3/(4u)$. Together with $g'(0) = 0$ they give $L'(0+) = 0$. Thus L is constant on the positive half. Reflection makes the same constant valid on the negative half, establishing stationarity against all real zero-mean perturbations.

Coercivity gives unique global minimality. At $s = 1/2$, evenness and normalization imply $g(1/2) = 1/2$. Multiplication of the Euler equation by f_u and integration show that its constant is $2F_u(f_u)$. Endpoint evaluation gives (8.2). Stationarity also proves, for every real normalized f,

$$F_u(f) = m(u) + F_u(f - f_u). \tag{8.6}$$

To prove positivity, write $x = s - 1/4$ on the positive half. For $x \ge 0$ both derivative terms in (8.5) are nonpositive, and $f_u(1/4 - x) \ge f_u(1/4 + x)$. The minimum is consequently

$$f_u(1/2) = \alpha\cot\alpha - \beta\tan\beta.$$

At $u = 1/8$ the arguments are 1 and $1/\sqrt{2}$. Alternating Taylor bounds give

$$\cos 1 \ge 389/720, \quad \sin 1 \le 101/120, \quad \frac{\sin(1/\sqrt{2})}{1/\sqrt{2}} \le 147/160, \quad \cos(1/\sqrt{2}) \ge 4379/5760.$$

Therefore

$$f_{1/8}(1/2) \ge \frac{389}{606} - \frac{2646}{4379} = \frac{99955}{2653674} > 0.$$

On $(0, \pi/2)$, $x\cot x$ decreases and $x\tan x$ increases. Increasing u decreases both arguments, so the endpoint remains positive. The signed affine minimum is thus also the minimum over nonnegative profiles.

8.3. **The frontier, its rigidity and the union witness.** The two lattice tests give nonnegative slacks

$$\sigma_1 = \overline{A} - 1 + d - a, \qquad \sigma_2 = \overline{A}/2 + \overline{C}_{1/2} - 1 + 2d - a.$$

Since $q\theta = 1 - 2u$ and $q(1-\theta) = u$, $q\sigma = (1-2u)\sigma_1 + u\sigma_2$. The general residual is therefore

$$\begin{aligned} \mathrm{Res} = m(u) - \frac{2u^2}{u-e} + \sum_i w_i F_u(f_i - f_u) \\ + \frac{u-e}{8} \sum_i w_i \left(\Gamma_i - \frac{4u}{u-e}\right)^2 + (1-2u)\sigma_1 + u\sigma_2. \end{aligned} \tag{8.7}$$

All terms after the first are nonnegative. The positive profile gives $m(u) > 0$, since $J(f_u) > 0$ and $3A(f_u)/2 - C_{1/2}(f_u) \geq A(f_u) > 0$. The exact sufficient frontier is

$$1/8 \leq u \leq 1/3, \quad 0 \leq e < u, \qquad e \leq E(u) := u - \frac{2u^2}{m(u)}. \tag{8.8}$$

The profile term costs at least $(u - 1/8 + 1/1000)\sum_i w_i \|f_i - f_u\|_2^2$. Near equality at the frontier controls the weighted squared departures from the same profile and threshold, together with the lattice-test slacks. Individual control requires a lower bound for that profile's weight.

Exactness refers to the enlarged dual problem constrained only by these two lattice tests. To verify it, use one profile f_u, write $E_1 = A(f_u) - 1$ and $E_2 = A(f_u)/2 + C_{1/2}(f_u) - 1$, and choose

$$d = E_1 - E_2 \geq 0, \qquad a = 2E_1 - E_2, \qquad \Gamma = 4u/(u-e).$$

Both lattice tests are equalities and every residual term after the first vanishes. No actual universal Gram floor with those coefficients has been asserted; other periodic states and clipping can impose further restrictions.

The function m is smooth by its formula and concave as an infimum of affine functions of u. Stationarity gives $m'(u) = 3A(f_u)/2 - C_{1/2}(f_u)$, so $m - um' = J(f_u) > 0$. Differentiation yields

$$\left(\frac{u^2}{m}\right)'' = \frac{2(m - um')^2}{m^3} - \frac{u^2 m''}{m^2} > 0.$$

Thus E is strictly concave on the stated interval. The following consequence needs no exact maximizer:

$$u_{\mathrm{U}} = \frac{13711914}{10^8}, \qquad e_{\mathrm{U}} = \frac{5251083389778}{10^{14}}, \qquad m(u_{\mathrm{U}}) > \frac{2u_{\mathrm{U}}^2}{u_{\mathrm{U}} - e_{\mathrm{U}}}. \tag{8.9}$$

For a finite rational verification, let $z = 1/(16u)$ and put

$$C_n(z) = \sum_{k=0}^{n} \frac{(-1)^k z^k}{(2k)!}, \qquad S_n(z) = \sum_{k=0}^{n} \frac{(-1)^k z^k}{(2k+1)!}.$$

For $0 < 2z \leq 1$, the index-15 sums bound the corresponding cosine and normalized sine below, and the index-16 sums bound them above; all denominator lower bounds are positive. It follows that

$$m(u) \geq m_L(u) := \frac{1}{2} + u\left\{\frac{C_{15}(2z)}{S_{16}(2z)} - 2z\frac{S_{16}(z)}{C_{15}(z)}\right\}.$$

Substitution of the rationals in (8.9) and positive-denominator cross multiplication give

$$m_L(u_{\mathrm{U}}) - \frac{2u_{\mathrm{U}}^2}{u_{\mathrm{U}} - e_{\mathrm{U}}} > 10^{-14}.$$

This specifies the certificate by finite rational expressions; no quadrature or optimization is a premise.

The union complement consists of multiple nonreal pairs. A pair of multiplicity $j \geq 2$ has excluded mass $2j$ and excess $2(j-1)$, so $N - N_{\mathrm{s}\cup 0} \leq 2eN$. At (8.9), equality is compatible with the count variables: assign mass $1 - u_{\mathrm{U}}$ to simple real locations, $u_{\mathrm{U}} - 2e_{\mathrm{U}}$ to simple nonreal pairs, and $2e_{\mathrm{U}}$

to double nonreal pairs. These masses are nonnegative. A double pair has mass four and excess two; multiplying by a common integer denominator gives integral populations with precisely the specified excess. The retained union relaxation therefore cannot force a lower proportion above

$$1 - 2e_{\mathrm{U}} = \frac{22374458305111}{25000000000000} = 0.89497833220444. \tag{8.10}$$

This is a feasible count population, not a construction of exponential vectors realizing every analytic energy.

The zero-excess endpoint is the root $r_* \in [8/25, 1/3]$ of $m(r_*) = 2r_*$. On that smaller interval $\alpha^2 \le 25/64$ and $\beta^2 \le 25/128$; elementary sine and cosine bounds imply

$$\|f_u\|_\infty \le \frac{384}{359} + \frac{50}{231} < \frac{4}{3}.$$

As $f_u \ge 0$ and $\int f_u = 1$, this gives $m'(u) < 2$. Thus $m(u) - 2u$ decreases strictly there. The same alternating expressions, with indices 9 and 10, give opposite signs at $32349386085/10^{11}$ and $32349386086/10^{11}$. Hence

$$0.67650613914 < 1 - r_* < 0.67650613915.$$

This is the simple-count endpoint of the two-lattice frontier; the next construction gives a stronger numerical obstruction.

9. A twelve-fibre obstruction for all profiles

Gebendorfer [13, Theorem 1.2 and Sections 5–6] used two periodic configurations to obtain the ceiling 0.674034 for a normalized nonnegative 25-mode profile family, with 24 free coefficients. The periodic configurations and their density balancing are his inputs. We solve the associated variational problem on the full affine L^2 space at the slightly larger value 0.67404. We also give a positive profile showing why the original finite-mode test cannot be extended unchanged.

9.1. The quadratic problem and its coercivity.

The two configurations are

$$A_{\mathrm{per}} : \ (0, \alpha_0) \pmod 3, \quad \alpha_0 = 5189/5000, \qquad B_{\mathrm{per}} : \ (0, \gamma_0, 2\gamma_0) \pmod 4, \quad \gamma_0 = 10289/10000.$$

By (7.10), their full per-point energies are

$$\begin{aligned} Q_A(f) &= \frac{1}{6}\left[4A(f) + 2\sum_{j=1}^{2}\{2 + 2\cos(2\pi j\alpha_0/3)\}C_{j/3}(f)\right], \\ Q_B(f) &= \frac{1}{12}\left[9A(f) + 2\sum_{j=1}^{3}\{3 + 4\cos(\pi j\gamma_0/2) + 2\cos(\pi j\gamma_0)\}C_{j/4}(f)\right]. \end{aligned} \tag{9.1}$$

For $2/3 < q < 3/4$, give the second configuration weight $\lambda_q = 9 - 6/q$. Then $(1-\lambda_q)3/2 + \lambda_q 4/3 = 1/q$. Writing $Q_q = (1 - \lambda_q)Q_A + \lambda_q Q_B$, we have

$$\mathcal{F}_q(f) = R(f) - qQ_q(f) = R(f) - (6 - 8q)Q_A(f) - (9q - 6)Q_B(f). \tag{9.2}$$

Every correlation shift is a multiple of $1/12$. Put $\ell = 1/12$, $x_j = -1/2 + j\ell$, and $F(x) = (f(x_0 + x), \dots, f(x_{11} + x))^t$ for $0 < x < \ell$. Let T_k have ones at $(j, j + k)$, $0 \le j < 12 - k$, and zeros elsewhere. These are truncated shifts, with no periodic identification of cell endpoints. If

$Q_q = c_0 A + \sum_{k\in\{3,4,6,8,9\}} c_k C_{k/12}$, then

$$\begin{aligned} c_0 &= (1-\lambda_q)\frac{2}{3} + \lambda_q \frac{3}{4}, \\ c_{4j} &= \frac{1-\lambda_q}{3}\{2 + 2\cos(2\pi j\alpha_0/3)\}, \qquad j = 1, 2, \\ c_{3j} &= \frac{\lambda_q}{6}\{3 + 4\cos(\pi j\gamma_0/2) + 2\cos(\pi j\gamma_0)\}, \qquad j = 1, 2, 3. \end{aligned} \tag{9.3}$$

The two index sets in the last lines are disjoint. Since $C_{k/12}(f) = \int_0^\ell F(x)^t T_k F(x)\,\mathrm{d}x$,

$$\begin{aligned} M_q &= (1 - qc_0)I - \frac{q}{2}\sum_{k\in\{3,4,6,8,9\}} c_k(T_k + T_k^t), \\ \mathcal{F}_q(f) &= \int_0^\ell F(x)^t M_q F(x)\,\mathrm{d}x + J(f). \end{aligned} \tag{9.4}$$

Set

$$q_{25} = \frac{337017}{500000}, \qquad q_{\mathrm{c}} = \frac{16851}{25000}. \tag{9.5}$$

The explicit rational enclosures in Appendix C give, at both values, every diagonal-dominance margin of M_q greater than $77/250$. Symmetry therefore gives $M_q > (77/250)I$. For a zero-mean perturbation, (8.4) and the Dirichlet Poincare inequality on the whole interval give $J(h) \geq -2\|h\|_2^2/\pi^2$. Consequently

$$\mathcal{F}_q(h) > \left(\frac{77}{250} - \frac{2}{(157/50)^2}\right)\|h\|_2^2 > \frac{1}{10}\|h\|_2^2 \qquad (h \neq 0,\ \int h = 0). \tag{9.6}$$

All forms are bounded on $L^2(I)$. Translate the affine space by any normalized function and restrict to the closed zero-mean subspace. The bounded positive operator representing the coercive bilinear form is invertible there, and its inverse solves the stationarity equation. Thus a unique affine minimizer exists. This controls the full space, with no finite-mode remainder omitted.

9.2. **A posteriori certification on the affine space.** For any normalized real candidate f_0, put $g_0(s) = \int_I |s-t| f_0(t)\,\mathrm{d}t$ and let

$$r = M_q f_0 + g_0 - c\mathbf{1}$$

for any real c, interpreting the multiplication in fibres. If $m_q = \min_{\int f = 1} \mathcal{F}_q(f)$, then

$$\mathcal{F}_q(f_0) - 10\|r\|_2^2 \leq m_q \leq \mathcal{F}_q(f_0). \tag{9.7}$$

Indeed, for $\int h = 0$,

$$\mathcal{F}_q(f_0 + h) = \mathcal{F}_q(f_0) + 2\langle r, h\rangle + \mathcal{F}_q(h) \geq \mathcal{F}_q(f_0) + 2\langle r, h\rangle + \tfrac{1}{10}\|h\|_2^2.$$

Completion of this Hilbert-space square proves the lower bound; the candidate itself proves the upper bound. The constant c disappears from the cross term because $\int h = 0$.

Appendix C defines one even, normalized, piecewise polynomial $f_0 > 7/10$ using 42 integer coefficients. Its evaluation is rational after seven cosine values have been enclosed by the displayed intervals. The resulting coarse enclosures, sufficient for the proof, are

$$\begin{aligned} \frac{651926252579}{10^{12}} &< \mathcal{F}_{q_{25}}(f_0) < \frac{651926252580}{10^{12}}, \\ \frac{651920180779}{10^{12}} &< \mathcal{F}_{q_{\mathrm{c}}}(f_0) < \frac{651920180780}{10^{12}}, \\ \left\| M_{q_{\mathrm{c}}} f_0 + g_0 - \frac{651920180779}{10^{12}}\mathbf{1} \right\|_\infty &< \frac{4}{10^{12}}. \end{aligned} \tag{9.8}$$

Since I has length one, the squared L^2 residual is less than $16/10^{24}$. It follows that

$$m_{q_c} > \frac{651920180779}{10^{12}} - \frac{160}{10^{24}} > 2 - 2q_c + \frac{18}{10^8}, \qquad 2 - 2q_c = \frac{8149}{12500}. \tag{9.9}$$

On the other hand the first line of (9.8) gives

$$\mathcal{F}_{q_{25}}(f_0) < 2 - 2q_{25} - \frac{1}{200000}. \tag{9.10}$$

Thus this periodic test does not exclude its original value over the larger profile space. This neither contradicts the source's finite-dimensional theorem nor excludes a different all-profile test at the 25-mode value.

9.3. **Positivity of the true minimizer.** The signed minimization already suffices for the obstruction. It is also the exact minimization over nonnegative profiles. Let $f^\dagger$ be the minimizer at q_c and put $c_0 = \mathcal{F}_{q_c}(f_0)$. The displayed enclosures give

$$\|M_{q_c} f_0 + g_0 - c_0 \mathbf{1}\|_\infty < 5 \cdot 10^{-12} < \varepsilon, \qquad \varepsilon = 10^{-11}.$$

Call this centered residual r_0 and set $h = f^\dagger - f_0$. The minimizing equation is $M_{q_c} f^\dagger + g_{f^\dagger} = m_{q_c}\mathbf{1}$: multiplication by $f^\dagger$ and integration identify its multiplier with the minimum. Subtract the candidate equation and pair with h to obtain $\mathcal{F}_{q_c}(h) = -\langle r_0, h\rangle$. Coercivity and (9.7) imply

$$\|h\|_2 \le 10\varepsilon, \qquad 0 \le c_0 - m_{q_c} \le 10\varepsilon^2.$$

Because $|s-t| \le 1$, $\|g_h\|_\infty \le \|h\|_2$. Strict row diagonal dominance gives $\|M_{q_c}^{-1}\|_{\ell^\infty \to \ell^\infty} < 250/77 < 4$: apply a row at a coordinate of maximum modulus in a finite-vector equation. The difference of the Euler equations therefore gives

$$\|h\|_\infty < 4(\varepsilon + 10\varepsilon + 10\varepsilon^2) < 10^{-9}.$$

The equation supplies this bounded representative fibre by fibre; boundedness was not presumed. Since $f_0 > 7/10$, $f^\dagger$ is positive almost everywhere. Reflection invariance and uniqueness also make it even.

For completeness there is an exact differential characterization, although certification did not require its numerical inversion. Put $G_j(x) = g_{f^\dagger}(x_j + x)$ and $V = G - m_{q_c}\mathbf{1}$. Then

$$V'' + 2M_{q_c}^{-1} V = 0, \qquad F = -M_{q_c}^{-1} V.$$

The 25 scalar conditions are continuity of G, G' at the eleven internal junctions, $G_0'(0) = -1$, $G_{11}'(\ell) = 1$, and $G_0(0) + G_{11}(\ell) = 1$. The endpoint derivatives enforce normalization through $g'' = 2f$. The endpoint sum follows from $|t + 1/2| + |1/2 - t| = 1$. Conversely these conditions identify G with the integral potential: their difference is globally affine, with zero endpoint derivatives and zero endpoint sum. Continuity is imposed on the potential and its derivative, not on the profile; the latter may jump at cell boundaries.

All forms in (7.1) are continuous in L^2: shifts are contractions, and $|s - t|$ is a bounded integral kernel on an interval of finite measure. The positive even f_0 and $f^\dagger$ can be approximated by profiles whose square roots are smooth and supported strictly inside I, as required in Section 2. For either f, first approximate $\sqrt{f}$ in L^4 by nonnegative even $\eta_j \in C_c^\infty((-1/2, 1/2))$, using symmetric cutoffs and even nonnegative mollifiers. The estimate

$$\|\eta_j^2 - f\|_2 \le \|\eta_j - \sqrt{f}\|_4 (\|\eta_j\|_4 + \|\sqrt{f}\|_4) \longrightarrow 0$$

shows that $\eta_j^2 / \int \eta_j^2 \to f$ in L^2. These normalized profiles have the required smooth square roots. The strict inequality (9.10) persists for a sufficiently close approximation. The variational infimum at q_c is unchanged on this dense admissible class, although its minimizer need not belong to the smooth class.

9.4. **The simultaneous method ceiling.** Apply (7.14) at the single count point $u = 1 - q_{\mathrm{c}}$, $e = 0$. Stationarity gives for every covered row

$$\begin{aligned} \mathrm{Res} &= \delta + \sum_i w_i \mathcal{F}_{q_{\mathrm{c}}}(f_i - f^\dagger) \\ &\quad + \frac{1 - q_{\mathrm{c}}}{8} \sum_i w_i (\Gamma_i - 4)^2 + q_{\mathrm{c}}\sigma, \qquad \delta = m_{q_{\mathrm{c}}} - (2 - 2q_{\mathrm{c}}) > \frac{18}{10^8}. \end{aligned} \tag{9.11}$$

The profile term is at least $\frac{1}{10}\sum_i w_i \|f_i - f^\dagger\|_2^2$. The same point therefore satisfies every joint-affine row, including arbitrary individual clipping thresholds. Its simple-real proportion is

$$q_{\mathrm{c}} = 0.67404, \qquad \text{giving the method ceiling } 67.404\%. \tag{9.12}$$

All omitted mass can be assigned to simple nonreal pairs. Every location is then simple and the distinct, simple-anywhere and union counts all equal N. Lower bounds for those statistics do not remove this count witness.

The comparison with earlier ceilings has two dimensions. Gebendorfer's smaller 0.674034 value applies on his finite-dimensional family; the present result covers every normalized L^2 profile in the specified joint-affine interface. Knausgård's fixed second-window bounds [10, Proposition 7.3] are smaller still: 0.67357 for seven index lags and 0.67383 for all pairs. His periodic-law result [10, Theorem 7.7 and Corollary 7.9], following [2, Section 7.2], treats a broader trace–energy construction with explicit regularity and variation costs. The present proof neither replaces every such construction nor identifies the optimal periodic mixture. It establishes the all-profile obstruction and quantitative residual for the rows defined in Section 7.

10. Interaction retained and information still available

The constructions above separate three uses of a quadratic zero energy. First, separation can make clipping exact on a large block. For the explicit cosine profile this retains the full local reward and yields (1.1). Second, local slack can control geometry rather than merely improve a scalar constant: the phase comparison bounds the departure of adjacent gaps from integral spacing, with an explicit price for arbitrarily long physical spans. Third, periodic configurations test the information content of affine deductions simultaneously over profiles and thresholds.

The variational results make the third distinction quantitative. Coercivity identifies the unique optimal profile for a prescribed periodic mixture and measures the cost of departing from it. The common residual identifies the independent cost of a nonoptimal threshold and the slack in the periodic tests. These are statements about the data retained by the deduction. They explain why further information about the same operator can matter even when an affine count has already been optimized.

The numerical proofs use two kinds of finite input. The eight-point floor for the main profile has full analytic enclosure and finite certificate data in the accompanying formal source. The labelled, third-profile and compensated-frame inequalities used elsewhere are the computer-assisted results cited at their statements; their published certificate computations are not replaced by the small arithmetic checks accompanying this paper. Those checks reproduce the displayed energy enclosures, table values, phase-polynomial bounds and variational residual certificates. In particular, the additional variational and multiplicity deductions are not asserted to be covered by the final exported statement of the supplied Lean file.

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Data availability

Numerical results reported here are reproducible from the verification scripts and certificate data in the repository. Lean code and verification details are also included. - Repository - github.com/arun-chandru/riemann-zeros.

Appendix A. The imported eight-point data and their algebraic consequences

This appendix specifies the local input for the second profile (6.1). The whole-domain inequality (6.3) is the computer-assisted input of [10, Proposition 5.3 and Tables 10–11], at $\theta = 17/20$ and $\alpha = 627/10^6$, with reward $a = 8722/10^6$. The tables below fix that premise completely. The finite capacity, potential, and slack calculations following them are separate elementary consequences. Listing the data is not a replay of the source's continuum search or its computation-dependent formalisation.

All entries in the next two tables are integer numerators with denominator 10^6. In the first table the row is r, the column is s, and the entry is $10^6 W_{rs}$; a dash is outside the range $0 \le r < s \le 7$.

$r\backslash s$	1	2	3	4	5	6	7
0	159939	0	280438	143766	709387	1000000	2000000
1	—	264496	385569	340591	856233	581225	1000000
2	—	—	389188	614430	757940	856233	709387
3	—	—	—	372750	614430	340591	143766
4	—	—	—	—	389188	385569	280438
5	—	—	—	—	—	264496	0
6	—	—	—	—	—	—	159939

The corrections are continuous piecewise-linear functions of a gap. Their values at the fourteen knots are as follows; all seven functions vanish outside $[3/4, 19/8]$.

t	$10^6 B_0(t)$	$10^6 B_1(t)$	$10^6 B_2(t)$	$10^6 B_3(t)$	$10^6 B_4(t)$	$10^6 B_5(t)$	$10^6 B_6(t)$
3/4	0	0	0	0	0	0	0
7/8	647	0	-425	-444	-425	0	647
1	555	91	-425	-442	-425	91	555
9/8	518	45	-342	-442	-342	45	518
5/4	647	0	-425	-444	-425	0	647
11/8	647	0	-425	-444	-425	0	647
3/2	647	0	-425	-444	-425	0	647
13/8	0	0	221	-442	221	0	0
7/4	0	0	221	-442	221	0	0
15/8	270	86	-275	-162	-275	86	270
2	325	58	-244	-278	-244	58	325
17/8	0	0	-425	850	-425	0	0
9/4	647	0	-425	-444	-425	0	647
19/8	0	0	0	0	0	0	0

Thus $(B_0, \dots, B_6) = (u_0, u_1, u_2, -2(u_0 + u_1 + u_2), u_2, u_1, u_0)$, where $u_j = B_j$ for $j = 0, 1, 2$. In particular $\sum_{r=0}^{6} B_r(t) = 0$ identically, including between knots.

The seven lag-capacity sums, in the same integer units, are

$$10^6 \sum_{r=0}^{7-h} W_{r,r+h} = (1999996, 1999998, 1999998, 1999998, 1999999, 2000000, 2000000)_h, \quad 1 \le h \le 7. \tag{A.1}$$

Every sum is at most $2 \cdot 10^6$. A pair in a translated configuration can occur at most once at each admissible relative position, so these are exactly the capacity bounds needed to charge its total weight to the ordered energy with coefficient two.

For completeness, the bounded potential can also be read directly from the table. Set $C_r = \sum_{s=0}^{r} B_s$ for $0 \le r \le 5$ and

$$\Phi(g_0, \dots, g_5) = \sum_{r=0}^{5} C_r(g_r).$$

The ranges of $10^6 C_r$ are respectively

$$[0, 647], \quad [0, 647], \quad [-425, 222], \quad [-222, 425], \quad [-647, 0], \quad [-647, 0].$$

These ranges include the zero values outside the knot interval. Linear interpolation cannot create a new extremum. Each range has length 647, and therefore

$$\operatorname{osc} \Phi \le \Omega := \frac{3882}{10^6}, \qquad \sum_{r=0}^{6} B_r(g_r) = \Phi(g_0, \dots, g_5) - \Phi(g_1, \dots, g_6). \tag{A.2}$$

On summing complete frames the potential telescopes. The spread sums to at most seven times the total spread, and the reward occurs $m - 7$ times. Keeping the nonnegative frame slacks gives exactly (6.4), with $B_0 = 7a + \Omega$; this scalar B_0 is distinct from the correction function $B_0(t)$ in the table. For $m < 8$ the same inequality holds with zero slack and empty sums.

The minima of the seven correction functions sum to $-(425 + 444 + 425)/10^6 = -1294/10^6$. Since all pair weights are nonnegative, each frame slack is consequently at least

$$\alpha(y_{i+7} - y_i) - a - 1294/10^6 = \alpha(y_{i+7} - y_i) - 10016/10^6.$$

This calculation supplies the span price used in the new argument and requires no positive margin for the source's accepted search boxes.

The associated separated-norm and close-pair inputs are also those of [10, Section 5]. Its majorant is

$$g(u) = \frac{13}{20} \left\{ \operatorname{sinc}\big(\theta(u - 1/2)\big) + \operatorname{sinc}\big(\theta(u + 1/2)\big) \right\}^2, \qquad \theta = 17/20.$$

The cited certificate proves $g \ge f$ on I and $K(t)^2 \ge 13707941/2^{28}$ on $|t| \le \theta$. The Fourier transform of g vanishes for $|\xi| \ge \theta$, and

$$\Lambda = \int_{\mathbb{R}} g = \frac{26}{17}(1 + \operatorname{sinc}(17/20)) < \frac{520}{289} < \frac{9}{5} < 2.$$

Here $\sin(17\pi/20) = \sin(3\pi/20) < 3\pi/20$. The support and integral follow directly from the sinc representation; the two continuum inequalities remain imported premises. These are the premises used in the smoothing and matching argument, rather than an inference about operator norms from entrywise approximation.

Appendix B. Exact arithmetic for the phase comparison and profile energies

The new finite calculations use rational arithmetic. This appendix gives the recurrences, endpoints, and inequalities needed to reproduce them without a floating-point grid or additional numerical data. The scalar checks verify consequences of the stated imported premises; they do not replace those premises.

B.1. **A common enclosure of π and the profile energies.** For $0 < x < 1$ put

$$A_n(x) = \sum_{k=0}^{n-1} \frac{(-1)^k x^{2k+1}}{2k+1}, \qquad r_n(x) = \frac{x^{2n+1}}{2n+1}.$$

For even n, the alternating-series remainder gives $A_n(x) < \arctan x < A_n(x) + r_n(x)$. Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, with $n = 24$ and $n = 8$ respectively, proves by rational comparison

$$\pi_- := \frac{3141592653589793}{10^{15}} < \pi < \pi_+ := \frac{3141592653589794}{10^{15}}. \tag{B.1}$$

Explicitly the inner lower endpoint is $16A_{24}(1/5) - 4\{A_8(1/239) + r_8(1/239)\}$ and the inner upper endpoint is $16\{A_{24}(1/5) + r_{24}(1/5)\} - 4A_8(1/239)$; they lie strictly between the displayed rational endpoints.

For the three integer-cosine profiles used above, write the energy as $A + B/\pi^2$ using (5.5). The exact pairs are

Profile	A	B
1	1624643573617/1200000000000	−250664498438947/1200000000000000
2	4026042764683/3000000000000	−440830631846157799/3175200000000000000
3	4060769086579/3000000000000	−1000828214586143/4800000000000000

All three values of B are negative. Thus the energy lies between $A + B/\pi_-^2$ and $A + B/\pi_+^2$, in that order. For convenient outward endpoints, multiplication by 10^{15} gives

$$\begin{aligned} 1332704958281710 &< 10^{15}R_1 < 1332704958281712, \\ 1327947272844555 &< 10^{15}R_2 < 1327947272844557, \\ 1332463633167058 &< 10^{15}R_3 < 1332463633167061. \end{aligned} \tag{B.2}$$

The first and third upper endpoints are below the caps in (5.6). The middle profile is used in the phase comparison. All three enclosures follow from the same short Machin calculation.

B.2. **The phase polynomial and its six rational intervals.** Use exactly the lags, radii, and weights in (6.14). Generate the Chebyshev coefficients by $T_0 = 1$, $T_1 = t$, $T_{l+1} = 2tT_l - T_{l-1}$. Form the rational numerator $N(t) = \sum_l w_l(1 - T_l(t)) - (1 - t)$ in (6.15). If $N(t) = \sum_{j=0}^{49} n_j t^j$, then $\sum_j n_j = 0$ and division by $1 - t$ gives

$$P(t) = \sum_{k=0}^{48} p_k t^k, \qquad p_k = \sum_{j=0}^{k} n_j.$$

The checks $n_{49} = -p_{48}$ and $P(1) = \sum_l w_l l^2 - 1$ provide endpoint identities independent of a sign test.

On any rational interval $[u, v]$, substitute $t = u + (v - u)x$. Its power coefficients and degree-48 Bernstein coefficients are

$$d_k = (v-u)^k \sum_{j=k}^{48} p_j \binom{j}{k} u^{j-k}, \qquad b_s = \sum_{k=0}^{s} d_k \frac{\binom{s}{k}}{\binom{48}{k}}, \quad 0 \le k, s \le 48. \tag{B.3}$$

The Bernstein basis is $\binom{48}{s}x^s(1-x)^{48-s}$; its entries are nonnegative and sum to one on $[0, 1]$. Direct use of (B.3) gives, for all 49 coefficients on each interval, the strict lower bounds in Table 2. Every bound exceeds $1/10000$. These 294 rational comparisons prove the phase inequality on the entire circle. Equivalently, start with the Bernstein coefficients on $[-1, 1]$ and apply de Casteljau midpoint subdivision to the six stated pieces; its updates are simply $(b_j + b_{j+1})/2$. The independently implemented check instead constructs T_l from the closed formula

$$T_l(t) = \frac{l}{2} \sum_{j=0}^{\lfloor l/2 \rfloor} \frac{(-1)^j (l-j-1)!}{j!(l-2j)!} (2t)^{l-2j} \quad (l \ge 1)$$

and uses direct interval conversion, avoiding the first implementation's recurrence and subdivision.

The other rigidity checks are short finite sums. With $c_j = (-1)^j a_j$ and $z_0 = 2601/25$, the tail lower bound is

$$1 + \sum_{c_j<0} \frac{c_j z_0}{z_0 - j^2} + \sum_{c_j>0} c_j = \frac{3123731672835001}{3750959784000000} > \frac{104}{125}.$$

For each lag the excess in the radius condition is exactly

$$R_l - \{q_l d_0/\alpha - (7q_l - l)\theta\} = \frac{16q_l}{627} > 0.$$

Using $\pi < 22/7$, the remaining rational comparisons are

$$\frac{196(22/7)^2}{8(104/125)^2} = \frac{1890625}{5408} < 350, \qquad \sum_{l=12}^{49} \frac{196q_l}{2\alpha R_l^3} < 112, \qquad \sum_{l=12}^{49} \frac{196l}{8R_l^2} < 6. \tag{B.4}$$

All denominators are positive. Together with the combinatorial proof in Section 6, these comparisons establish the displayed energy, slack, and boundary prices.

B.3. **Primitive pressure values and scalar guards.** Integer derivatives can be reconstructed without trigonometric interval arithmetic. Put $s(0) = 0$ and $s(n) = (-1)^n/n$ for nonzero integers. The sinc expansion gives

$$K'(k) = s(k) + \frac{1}{2}\sum_{j=1}^{10} a_j\{s(k-j) + s(k+j)\}, \qquad K(k) = a_k/2 \quad (k = 1, 2).$$

This yields the signed values in Section 6.4. The profile absolute-coefficient sum is $210672/10^6 < 1$, and the second-derivative bound has the explicit margin

$$4 - \frac{(22/7)^2}{3}\left(1 + \frac{210672}{10^6}\right) = \frac{31793}{2296875} > 0.$$

For $A_k = |K(k)|$, $B_k = |K'(k)|$, $r_1 = 1/20$, $r_2 = 1/25$, and the fixed p, c, C in (6.18), the primitive inner and outer guards are

Quantity	$k = 1$	$k = 2$
$A_k - B_k r_k - 2r_k^2$	36124241/5280000000	395330317/154000000000
$q_0 = 2A_k^2 + pk - c$	45448441/20000000000	7255009/125000000000
$B_k - 2r_k$	233921359/264000000	2084993683/6160000000
$2Cr_k - p$	1883/4000	1503/4000
$p(k - r_k) + Cr_k^2 - c$	2583/400000	259/40000

They are all positive. The quadratic coefficients are

$$q_2 = \frac{216248962271606881}{348480000000000000} > 6 \quad (k = 1), \qquad q_2 = \frac{93349805577714704489}{18972800000000000000} > 4 \quad (k = 2).$$

The two exact discriminants are those in (6.19). The left-interval slopes $4A_k(B_k - 2r_k) - p$ are the positive fractions printed there in the proof, and $5p/2 - c = 117/100000 > 0$ pays every remaining nearest integer. Thus the rational checks cover the Taylor positivity, both quadratic minima, the left intervals, outer monotonicity, and the unbounded range of gaps.

The affine and smoothing arithmetic is

$$Q = 1 + 350(19/4) = 3327/2,$$
$$B_\star = \frac{350(19/4)B_0 + c + 6(19/4)}{Q} = \frac{27293111}{332700000},$$
$$\frac{350(19/4)270 + 6}{Q} = \frac{299254}{1109}.$$

The pair and span guards can be read without decimal approximation:

$$0 < a_\star = \frac{725489}{83175000} < \frac{9}{1000} < \frac{1}{20} < \frac{13707941}{2^{28}}, \qquad 0 < b_\star = \frac{584077}{133080000} < \frac{4389}{10^6}.$$

They also imply $1 - a_\star > 0$. These are the guards used to pass from the separated finite certificate to the global count.

Appendix C. Exact data for the twelve-fibre certificate

This appendix specifies all arithmetic used in (9.8): finite rational polynomial integration and elementary transcendental enclosures. A numerical boundary-value solve helped find the candidate, but plays no part in the certificate.

C.1. **The normalized positive candidate.** On cell j put $y = 24(s - x_j) - 1 \in [-1, 1]$. For $0 \le j \le 5$ define $p_j(y) = 10^{-14} \sum_{k=0}^{6} n_{jk} y^k$. The integer data are split into two tables:

j	n_{j0}	n_{j1}	n_{j2}	n_{j3}
0	81452785069661	7292373590683	-495291728588	-8832261778
1	94237762244332	5325472480437	-540048842503	-5392125542
2	102829051521759	3211889787491	-561973129360	-1752444969
3	106611512235993	1220224746606	-555948929227	1548835301
4	105872959612158	3026144941748	-520652451189	-4834131202
5	110067398521096	987168788008	-542216720202	-1573291099

j	n_{j4}	n_{j5}	n_{j6}
0	468777856	3579876	-172336
1	498965786	1817587	-181576
2	508473653	-36353	-183294
3	491383198	-1705511	-175034
4	443772398	2367437	-154759
5	463005280	777705	-161751

Define $p_{11-j}(y) = p_j(-y)$ and divide the full function by

$$Z_0 = \frac{62999999999997821}{63000000000000000}. \tag{C.1}$$

Thus $f_0(s) = p_j(y)/Z_0$ on cell j. Using $ds = dy/24$ and $\int_{-1}^{1} y^k \, \mathrm{d}y = 2/(k+1)$ for even k, zero for odd k, gives exactly Z_0 as the unnormalized mass. Reflection gives evenness. On each cell the constant coefficient of p_j/Z_0, less the sum of the absolute values of its other coefficients, exceeds $7/10$. This proves uniform positivity on the whole cell.

All remaining polynomial operations can be specified by two linear maps. If $v(y) = \sum_k v_k y^k$, put

$$\mathcal{I}(v) = \frac{1}{12} \sum_{\substack{k \ge 0 \\ k \text{ even}}} \frac{v_k}{k+1}, \qquad \mathcal{P}(v)(y) = \sum_{k \ge 0} \frac{v_k}{k+1} \{ y^{k+1} - (-1)^{k+1} \}.$$

Here $\mathcal{I}$ is the physical-cell integral and $\mathcal{P}(v)(-1) = 0$. Write $v_j = p_j/Z_0$. Starting with $\mu_0 = 0$ and $\nu_0 = 1/2$, define

$$\begin{aligned} P_j(y) &= \mu_j + \mathcal{P}(v_j)(y)/24, \\ g_j(y) &= \nu_j + \mathcal{P}(2P_j - 1)(y)/24, \\ \mu_{j+1} &= P_j(1), \qquad \nu_{j+1} = g_j(1). \end{aligned} \tag{C.2}$$

The initial potential is $1/2$ by evenness and normalization. The recursion gives the cumulative integral and exact potential; $\mu_{12} = 1$, $\nu_{12} = 1/2$, and

$$\begin{aligned} A(f_0) &= \sum_j \mathcal{I}(v_j^2), \\ C_{k/12}(f_0) &= \sum_{j=0}^{11-k} \mathcal{I}(v_j v_{j+k}), \\ J(f_0) &= \sum_j \mathcal{I}(v_j g_j) = 2\sum_j \mathcal{I}(P_j - P_j^2). \end{aligned} \tag{C.3}$$

The last equality supplies a consistency check between two exact integrations.

C.2. **Seven explicit cosine enclosures.** Every transcendental entry in (9.3) is one of the seven values below. The integer L_r in the second column satisfies

$$\frac{L_r}{10^{18}} < \cos(\pi r) < \frac{L_r + 1}{10^{18}}. \tag{C.4}$$

r	L_r
5189/7500	−566923938842617133
5189/3750	−357194495134345020
10289/20000	−45380423447917089
10289/10000	−995881234335375474
30867/20000	135767447683864460
10289/5000	983558865802702074
30867/10000	−963134400298818254

To verify the table, for a positive integer b put $A_n(b) = \sum_{k=0}^{n} (-1)^k/((2k+1)b^{2k+1})$. The alternating remainder and Machin's identity imply

$$16A_{65}(5) - 4A_{20}(239) < \pi < 16A_{64}(5) - 4A_{21}(239).$$

Reduce each rational multiplier modulo two and by reflection to $\widetilde{r} \in [0, 1]$. Multiplication by the displayed interval for π gives $X = [X_-, X_+]$. Evaluate $\sum_{k=0}^{40} (-1)^k X^{2k}/(2k)!$ using rational interval arithmetic and enlarge both endpoints by $X_+^{82}/82!$. Taylor's theorem to degree 81 justifies this enlargement, since all real derivatives of cosine have absolute value at most one. The resulting interval lies strictly inside the corresponding interval in (C.4). Thus the seven entries follow from the displayed finite sums.

C.3. **Matrix and residual acceptance inequalities.** Insert (C.4) into (9.3)–(9.4) with outward rational interval arithmetic. The diagonal is exact: $1 - qc_0 = 3/2 - 17q/12$. There are three row-diagonal-dominance types. The last two columns below are strict lower bounds with denominator 10^{12}:

Row indices	$q = q_{25}$	$q = q_{\mathrm{c}}$
$0, 1, 2, 9, 10, 11$	308234221665	308227513777
$3, 8$	312599837175	312596684670
$4, 5, 6, 7$	355085988964	355079480788

Each exceeds $77/250$. The calculation is the exact diagonal less the sum of the absolute upper bounds of the other entries.

Use (C.3) in (9.2) for the energies. For the residual at q_{c} take $c = 651920180779/10^{12}$ and form on each cell

$$r_j(y) = \sum_{k=0}^{11} (M_{q_{\mathrm{c}}})_{jk} v_k(y) + g_j(y) - c.$$

If its coefficient intervals are $[a_{jr}, b_{jr}]$, put $V_j = \sum_r \max(|a_{jr}|, |b_{jr}|)$. Then $|r_j(y)| \le V_j$ on $[-1,1]$ and $\|r\|_2^2 \le \sum_j V_j^2/12$. The rational data give

$$651926252579 < 10^{12}\mathcal{F}_{q_{25}}(f_0) < 651926252580,$$
$$651920180779 < 10^{12}\mathcal{F}_{q_c}(f_0) < 651920180780,$$
$$V_j < 4/10^{12} \qquad (0 \le j \le 11).$$

These are the coarse certificates in (9.8). Every integral, coefficient and comparison is specified above as a rational operation on displayed integers or the seven proved intervals. The residual estimate (9.7) turns these finite inequalities into the all-L^2 conclusion.

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